

AT-FSV / ATAI
CMFSB ISOLATION MODULE — DOCUMENT 00
MASTER ENGINEERING BASELINE | ATFSV-MOD-001 | Captured Mechanical Floating Safety Body
Revision 5.16 — Single-Layer Controlled Baseline for Licence and IP-Transaction Negotiation
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DOCUMENT CONTROL

Field	Controlled value
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Controlled limitation. This document supports T1 technical diligence, supplier evaluation, licence negotiation and IP-transaction negotiation. It is not a performance certificate, type approval, safety case, measured ride-comfort claim, production commitment, warranty, patent grant, price list or regulatory approval.

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0. REVISION AND SUPERSESSSION

- 0.1 Identity rule. One document ID identifies exactly one content. All earlier files that carry IDs R5.11–R5.15 are WITHDRAWN and are not active design inputs. Only values in Section 3 and requirement text in Section 7 are active.
- 0.2 Corrective items and engineering status. "Applied" means corrective text is in this document. It does not mean engineering closure.

ID	Topic	Applied in	Engineering status
C-01	LOCKED geometry is not packaging closure	\$4	OPEN — V-L residual –120 mm
C-02	6.3 kW is Lorentz copper loss only	\$5	OPEN — ENE-001
C-03	48 V current budget	\$5	OPEN — ENE-001 / ACT-001
C-04	REQ-M07 quantitative criteria	\$7.2	OPEN — limits not frozen
C-05	REQ-M15 under power and hydraulic loss	\$7.3	OPEN — physical trial
C-06	REQ-M21 full sensor/fault/transient policy	\$7.4	OPEN — SIM/HIL evidence
C-07	Sensor functional independence	\$7.5	OPEN — SAF-001
C-08	Km evidence chain	\$10.1	OPEN — ACT-001 map
C-09	Gap tolerance envelope	\$10.2	OPEN — ACT-001
C-10	Progressive-stop curve	\$7.6	OPEN — STR-001
C-11	REQ-M25 horizontal path	\$7.7	OPEN — STR-001 / ICD-001
C-12	Multi-family IP, ownership, disclosure	\$16, Annex T3-A	OPEN — filing status below
C-13	Tower mass statistical basis	\$6.1	OPEN — supplier data
C-14	Static floor rule, rounding and traceability	\$6.2	Applied; inputs STUDY
C-15	Stack-up arithmetic and missing lines	\$4	OPEN
C-16	State machine gaps (S2/S2S→S3)	\$8	Applied; verification OPEN
C-17	"No time filter" removed from REQ-M21	\$7.4	Applied; parameters TBD
C-18	Mode-time allocation	\$7.1, \$11	Applied; values TBD

1. PURPOSE, NON-CLAIMS, VALUE AND DISCLOSURE TIERS

- 1.1 Purpose. This baseline defines ATFSV-MOD-001: a mechanically captured, dual-mode isolation module for specialty, defence-utility, ambulance and limited-series vehicles. It fixes architecture, requirements, evidence plan, risks, interfaces, host references and the transaction boundary.

1.2 Non-claims register (single location).

#	Non-claim
1	No production readiness or series manufacturing capability
2	No regulatory approval, type approval or certification credit
3	No certified ASIL or equivalent safety integrity level
4	No measured ride-comfort percentage or guaranteed comfort improvement until L1/L2 evidence
5	No crashworthiness or survivability demonstration
6	No guaranteed vehicle mass compliance (4,000 kg / 7,000 kg ceilings are engagement targets)
7	No final supplier mass, cost or lead time
8	No closed mass or energy budget suitable for production release
9	No continuous public-road 250 km/h capability and no continuous RMS force capability until thermal bench data exist
10	No clinical outcome claim for ambulance use
11	No patent grant, novelty, enforceability or freedom-to-operate conclusion before counsel-led analysis
12	No FIA homologation or series eligibility from FIA-referenced design philosophy
13	No supplier or OEM endorsement
14	No occupant-escape or detachable-capsule system is part of this module baseline
15	No warranty, price list or offer of exclusive sale by this document

Contradictory external presentations shall be withdrawn. Sales language may describe architecture and modes and may not invent numbers.

1.3 Value statement. Occupant compartments in demanding vehicles receive road vibration that affects comfort, task performance and, where relevant, clinical stability. Soft isolation alone reduces handling stiffness in rapid manoeuvres. CMFSB combines a COMFORT isolation envelope with a SPORT mechanical lock, keeps the upper body mechanically captured at all times, and settles on passive support on power loss. Measurable advantage is not claimed until L1/L2 evidence exists.

1.4 Planning KPIs (method definitions only).

KPI	Metric definition	Method	Status
Ride isolation	aw (0.5–80 Hz), VDV, MSDV (0.1–0.5 Hz) per ISO 2631-1	SIM-001, then bench	OPEN until L1
Handling stiffness	Roll angle and lateral response, SPORT vs COMFORT, same step	SIM-001, mule	OPEN until L1
Mode time	PARAM-MODE-TIME, §7.1 definition	Bench / HIL	TARGET
Power loss	S4 with quantitative envelope (§7.2)	Power-cut test	BASELINE
Module mass	PARAM-TOWER-MASS-LIMIT at PARAM-MASS-CONFIDENCE	Weighing	TARGET

1.5 Disclosure tiers.

Tier	Content	Access
T0	Public non-enabling concept / teaser	Public
T1	This baseline: architecture, requirements, risk, verification planning	NDA
T2	Models, detailed ICD, test data, configuration-controlled design artefacts	NDA + need-to-know
T3	Claim charts, prior-art mapping, IP schedule, negotiation strategy, commercial terms (COM-001)	NDA + counsel + restricted list

Disclosure is minimised to the lowest tier required. Detailed actuator geometry, control parameters and claim-sensitive implementation stay in T2/T3.

1.6 Status vocabulary.

Label	Meaning
BASELINE	Architectural decision for this revision; verification still required
TARGET	Numeric design goal; not verified
STUDY	Indicative planning figure; may change
ASSUMPTION	Input without programme-owned evidence
DERIVED	Calculated from stated inputs; inherits their uncertainty
LOCKED	Programme engagement decision; not test verification and not packaging closure
OPEN / TBD	Unresolved at this revision
OPTION	Not committed
SUPERSEDED / HISTORICAL	Retained for traceability only

2. ARCHITECTURE, SAFETY INVARIANTS AND MODES

2.1 Architecture. The upper body is kinematically constrained at all times (no free levitation). Four corner towers each combine: a hydraulic macro stage (PARAM-M02-MACRO-STROKE) with leveling and lock seats; a Lorentz micro stage (PARAM-M03-MICRO-STROKE) within the PARAM-LORENTZ-GAP envelope; progressive mechanical stops; and an optional SPORT positive lock. A shared ECU performs mode arbitration, the REQ-M21 interlock and power-loss detection. Layer stack, top to bottom: upper occupant/safety body → micro stage → macro stage → lower chassis. Architectural signature: the body is always mechanically captured; active isolation operates inside a bounded mechanical safety envelope; comfort control, handling stiffness and crash containment are coordinated but independently bounded. This signature stays valid if the Lorentz micro stage is replaced by another high-bandwidth actuator.

2.2 Independent load and force paths.

Path	Carrier
Normal vertical load	Passive / compliant mechanical-hydraulic support
Comfort dynamic force	Lorentz micro stage (force-bounded)
Large displacement and leveling	Hydraulic macro stage
Crash / abnormal load	Mechanical capture + progressive stops (independent of Lorentz closed loop)
Horizontal load	Capture frame, links, horizontal stops, SPORT lock faces (§7.7)
Power loss	S4 passive support with quantitative envelope (§7.2)

2.3 Safety invariants.

Invariant	Rule
Mechanical capture	No operating state permits loss of geometric capture
Static support	Vehicle weight on passive or fail-safe compliant paths, never solely magnetic
Power loss	S4 passive support; valves fail-closed; software aliveness not required

Invariant	Rule
Crash path	Progressive stops and structure independent of closed-loop magnetic force
Mode transition	No transition enters unconstrained free-float
SPORT lock fault	Lock failure shall not defeat capture or progressive stops
Horizontal restraint	Dedicated paths and/or SPORT lock; REQ-M25 closes before any integration-ready claim
Override limit	Driver override cannot defeat the REQ-M21 interlock
Manual rescue	Two-step mechanism operable under power loss and hydraulic failure; no unintended free-float

2.4 Modes.

Mode	State	Mechanical condition	Objective
COMFORT	S2	Locks released; macro and micro active; leveling active	Soft ride isolation
SPORT	S2S	Upper body seated; all required locks confirmed; stages held	Handling stiffness

Selection inputs: driver command (highest priority), radar/surface proposal, GPS/map proposal. Default after self-test is COMFORT unless ICD-001 states otherwise. Hold function (optional): rate-limited hydraulic orifice or friction device at the macro stage; not a primary crash path; velocity and force limits are frozen in ICD-001; if not fitted, REQ-M16 is N/A.

2.5 Dynamics screening (STUDY). With PARAM-REF-MASS-OCC and four corners, corner share is 662.5 kg; with PARAM-K-EFF the natural frequency is PARAM-FN-STUDY. A payload change of 2.4 kN gives about 60 mm at PARAM-K-EFF, so active leveling (REQ-M22) is required. Static sag is carried by gas-spring preload and leveling, not by the dynamic stroke. Carriage first structural mode shall be at least three times the micro control bandwidth.

3. MASTER PARAMETER REGISTER (one value, one place)

Other sections cite the parameter ID and do not restate the value.

Parameter ID	Value	Status
PARAM-M02-MACRO-STROKE	±50 mm body-relative (100 mm range)	TARGET
PARAM-M03-MICRO-STROKE	±7.5 mm	TARGET
PARAM-LORENTZ-GAP	2–5 mm tolerance-controlled envelope (15–30 mm applies to PM+EM OPTION only)	TARGET
PARAM-FORCE-PEAK	±2.5 kN per tower	STUDY
PARAM-FORCE-RMS	≈0.7 kN per tower	STUDY
PARAM-ACT-KM-MIN	≥63 N/√W	ASSUMPTION until ACT-001
PARAM-COPPER-MEAN	≈123 W per tower; ≈0.5 kW four towers (limit ≤0.5 kW) at PARAM-FORCE-RMS	DERIVED / TARGET
PARAM-COPPER-PEAK	≈1.57 kW per tower; ≈6.3 kW four towers; copper loss only	DERIVED
PARAM-SUPPLY-NOM	48 V class	STUDY
PARAM-BW-CONTROL	≥20 Hz; carriage first mode ≥3×	TARGET
PARAM-K-EFF	≈40 kN/m about the leveled operating point	STUDY
PARAM-FN-STUDY	≈1.23 Hz	DERIVED / STUDY
PARAM-REF-MASS-OCC	≈2,650 kg occupied upper body	STUDY
PARAM-REF-MASS-MAX	≈3,150 kg maximum-study upper body	STUDY
PARAM-TOWER-MASS-P50	48 kg	STUDY
PARAM-TOWER-MASS-LIMIT	≤55 kg	TARGET at PARAM-MASS-CONFIDENCE
PARAM-MASS-CONFIDENCE	TBD in CFG-001 before G1 (candidate P90)	OPEN
PARAM-TOWER-MASS-P90-LINESUM	67 kg (conservative line-sum of element P90 values)	STUDY
PARAM-STATIC-RULE	Floor = round-up to 0.5 kN of 1.25 × maximum static corner load	BASELINE
PARAM-STR-S-STATIC-FLOOR	≥5.0 kN per corner	TARGET
PARAM-STR-M-STATIC-FLOOR	≥7.0 kN per corner	TARGET
PARAM-STR-L-STATIC-FLOOR	≥10.0 kN per corner	TARGET (9.7 kN is SUPERSEDED)
PARAM-M21-AY-HIGH	0.35 g	Temporary TARGET
PARAM-M21-AY-LOW	0.30 g	Temporary TARGET
PARAM-M21-HOLD	TBD (SIM-001 / HIL)	OPEN
PARAM-ROLL-GRADIENT-VL	≤6.0 deg/g at V-L design payload	TARGET
PARAM-MODE-TIME	≤3 s, accepted request to declared state, excluding REQ-M21 inhibit time	TARGET
PARAM-LIFE	≥10,000 h equivalent road duty	TARGET
PARAM-OCC-MASS	100 kg per occupant	Policy
PARAM-GVW-VL	4,000 kg ceiling	LOCKED (engagement)
PARAM-GVW-VH	7,000 kg ceiling	LOCKED (engagement)
PARAM-REVIEWER-DEADLINE	Earlier of 2026-12-05 and G1 closure	Governance

4. PACKAGING FEASIBILITY GATE

4.1 Rule. LOCKED vehicle envelope values (overall height, ground clearance and related host geometry in §14) are engagement targets and are not packaging closure. Until H-point, cabin height, tower integration, macro travel, free play, stop stroke, micro-stage clearance and chassis integration are configuration-controlled, geometry remains under this Gate. No production or integration claim is made until the complete vertical and horizontal stack-up closes with a non-negative verified margin. Do not force a non-negative residual by reducing cabin height below H-point requirements.

4.2 Vertical stack-up (SPORT / seated).

Element	V-L (mm)	V-H (mm)	Status
Overall height	1,700	2,000	LOCKED (engagement)
Ground clearance	200	300	LOCKED (engagement)
Available for structure	1,500	1,700	DERIVED
Chassis / axle structure	280	320	STUDY
Tower structure and mounts	120	140	STUDY
Macro travel (100) + free play (40)	140	140	TARGET + STUDY

Element	V-L (mm)	V-H (mm)	Status
Cabin interior height from H-point	≥1,000	≥1,000	TARGET
Roof / exterior shell	80	90	STUDY
Allocated sum	≥1,620	≥1,690	DERIVED
Residual margin (before lines below)	≤ -120	≤ +10	OPEN — P0
Stop stroke (STR-001)	not yet in stack	not yet in stack	OPEN
Micro-stage clearance (PARAM-M03-MICRO-STROKE)	not yet in stack	not yet in stack	OPEN
Tolerance, thermal and static-sag allowance	not yet in stack	not yet in stack	OPEN

V-H residual is exactly +10 mm with the stated numbers; any stop stroke above 10 mm makes it negative unless the stroke is shown to lie inside the tower allowance. Closure options: integrate tower structure inside chassis depth; raise overall height; reduce ground clearance; package free play in the underfloor.

5. POWER AND CURRENT ENVELOPE

5.1 Rule. PARAM-COPPER-PEAK is Lorentz copper loss at simultaneous peak force on four towers. It is not module electrical demand. Electrical input power is copper loss plus mechanical power plus driver/converter losses. Total system power shall include: Lorentz actuators; hydraulic pump; valve actuation; lock actuation; ECU and sensors; transient and inrush; driver/converter losses; safety and thermal derating margins.

5.2 Current. PARAM-COPPER-PEAK divided by 48 V gives ≈131 A, an indicative lower bound for bus current, not a design current. ENE-001 and ACT-001 shall define: continuous, RMS and peak current; peak duration; simultaneous-tower demand; driver efficiency; bus voltage tolerance; cable and connector losses; fuse/protection strategy; thermal derating; transient and regenerative conditions.

5.3 Power budget.

Consumer	Mean (W)	Peak (W)	Closure
Lorentz, four towers	≤500	≈6,300 (copper only)	ACT-001
Hydraulic pump	OPEN	OPEN	HYD-001
Valves	OPEN	OPEN	HYD-001
Lock actuators	OPEN	OPEN	LCK-001
ECU and sensors	OPEN	OPEN	ICD-001
Total system	OPEN	OPEN	ENE-001

6. MASS, STATIC FLOORS AND LOAD CASES

6.1 Module mass and statistical basis. PARAM-TOWER-MASS-LIMIT applies to the core tower only. If it is a production design limit, compliance shall be shown at PARAM-MASS-CONFIDENCE, not at P50. P50 is not "agreed confidence". PARAM-TOWER-MASS-P90-LINESUM is a line-sum of element P90 values and is a conservative upper bound; aggregate P90 shall be computed from element distributions (RSS or Monte-Carlo) when supplier data exist. Risk R-01 tracks this.

Element	P50 kg	P90 kg	Scope
Local structure per tower	18	25	Core
Lorentz and local carriage	22	30	Core
Sensors and local electronics	8	12	Core
Core tower subtotal	48	67	Limit TARGET; P90 under R-01
Cylinder and accumulator share	15	22	Corner
Valves, hoses, seals	5	8	Corner
Progressive stops	8	12	Corner
SPORT lock share	6	10	Corner
Harness, dampers, bellows	4	6	Corner
Pump and reservoir share (¼)	8	12	Shared
Full corner, Class L	≈94	≈137	STUDY
Four corners, Class L	≈376	≈548	Vehicle WBS isolation line
Per-class corner P50 (planning)	S ≈70 · M ≈85 · L ≈95		M interpolated until supplier data

6.2 Static support floor. Rule: PARAM-STATIC-RULE. Mass basis for each class is stated so every floor is traceable.

Class	Mass basis	Maximum static per corner	1.25 × max	Floor (0.5 kN round-up)
S (V-L)	Occupied corner share 310–350 kg gives 3.0–3.4 kN; planning envelope with payload and margin	≈4.0 kN (STUDY)	5.0	≥5.0 kN
M (intermediate)	Corner share 450–550 kg gives 4.4–5.4 kN	≈5.5 kN (STUDY)	6.88	≥7.0 kN
L (V-H / generic)	PARAM-REF-MASS-MAX ÷ 4 × 9.81	≈7.7 kN (7.73)	9.66	≥10.0 kN

Class L inputs come from the generic maximum-study reference mass, which is more conservative than the V-H budget in §14.4 (upper body about 2,220–2,420 kg, ≈5.4–5.9 kN per corner). The generic reference is retained deliberately and is not derived from §14.

6.3 Reference load cases. Class L is shown on both mass bases; crash cases shall state which basis is the design case in STR-001.

Case	Class S	Class M	Class L (PARAM-REF-MASS-OCC)	Class L (PARAM-REF-MASS-MAX)
Corner mass share	310–350 kg	450–550 kg	662.5 kg	787.5 kg
3 g operational per tower	8–11 kN	12–17 kN	20–23 kN	≈23 kN
20 g vertical per tower	60–70 kN	90–110 kN	≈130 kN	≈154.5 kN
×2 asymmetry per tower	120–140 kN	180–220 kN	≈260 kN	≈309 kN
Horizontal 1 g, total	12–14 kN	18–22 kN	≈26 kN	≈30.9 kN
Horizontal 20 g, total	250–280 kN	360–440 kN	≈520 kN	≈618 kN

All values STUDY. Risk R-08 compares crash loads with capture frame, stops and hardpoint capacity.

6.4 Stop-energy illustration. A placeholder of ≈15.6 kJ results from $\frac{1}{2} m v^2$ with $m = \text{PARAM-REF-MASS-OCC}$ and $v = 3.43 \text{ m/s}$ (20 g over 30 mm free play). It is illustrative. STR-001 shall define the design energy event: initiating scenario, relative velocity, free-play geometry, corner versus whole-body share and mass case, and shall replace this value.

7. CONTROLLED REQUIREMENTS

7.1 Requirement table.

ID	Requirement	Owner	Gate	Status
REQ-M01	Upper body mechanically captured in all states; static capacity per class S/M/L at the §6.2 floors	Structures	G0/G2	BASELINE / TARGET
REQ-M02	Macro stroke PARAM-M02-MACRO-STROKE without clash	Mechanical	G1	TARGET
REQ-M03	Micro stroke PARAM-M03-MICRO-STROKE inside PARAM-LORENTZ-GAP envelope	Actuator	G1	TARGET
REQ-M04	Control bandwidth and carriage mode per PARAM-BW-CONTROL	Control	G2	TARGET
REQ-M05	Peak and RMS force per PARAM-FORCE-PEAK / PARAM-FORCE-RMS; Km measured, PARAM-ACT-KM-MIN stays ASSUMPTION until ACT-001	Actuator	G2	ASSUMPTION
REQ-M06	Core tower mass per PARAM-TOWER-MASS-LIMIT at PARAM-MASS-CONFIDENCE	Module	G2	TARGET
REQ-M07	Power-loss quantitative criteria (§7.2)	Safety	G2	BASELINE
REQ-M08	SIM-001 multi-band acceptance (§12.2)	Simulation	G1	TARGET
REQ-M09a	STR-001 per-class progressive-stop curve including §6.3 Class L cases	Structures	G3	TARGET
REQ-M09b	Stops absorb crash energy without the magnet loop; full curve (§7.6)	Structures	G3	TARGET
REQ-M10	Life per PARAM-LIFE	Reliability	G3	TARGET
REQ-M11	Operating temperature range frozen	Thermal	G2	OPEN
REQ-M12	EMI and cabin-field plan	EMC	G3	OPEN
REQ-M13	ICD-001 defines mechanical, electrical, communication, safety and lock interfaces	Systems	G1	TARGET
REQ-M14	No single failure causes loss of mechanical capture	Safety	G2	TARGET
REQ-M15	Two-step manual rescue under power and hydraulic loss (§7.3)	Safety	G3	TARGET
REQ-M16	If hold is fitted: rate-limited; not a primary crash path	Safety	G2	BASELINE (conditional)
REQ-M17	Required SPORT locks positively confirm engagement and release; time is an allocation inside PARAM-MODE-TIME (value TBD in ICD-001)	Lock	G2	TARGET
REQ-M18	COMFORT provides full macro envelope without lock interference	Mechanical	G2	TARGET
REQ-M19	Mode arbitration supports driver, radar/surface, GPS/map with driver priority	Controls	G3	STUDY
REQ-M20	No transition enters unconstrained free-float	Safety	G2	BASELINE
REQ-M21	SPORT↔COMFORT lateral-g policy (§7.4)	Dynamics	G2	TARGET
REQ-M22	Leveling preserves dynamic stroke and design attitude under payload change	Hydraulics	G2	TARGET
REQ-M23	Incomplete lock set inhibits SPORT declaration; lock state diagnosable and reported to ECU	Lock / Controls	G2	TARGET
REQ-M24	Fluid systems follow ISO 4413 safety path; interfaces in ICD-001	Hydraulics	G1	TARGET
REQ-M25	Horizontal load path defined and verified before integration-ready claim (§7.7)	Structures	G2	OPEN

PARAM-MODE-TIME is measured from accepted request to declared state (S2S or S2) and excludes REQ-M21 inhibit time.

7.2 REQ-M07 — Quantitative power loss. After loss of commanded active power the system shall: (1) detect it within a defined maximum detection time; (2) transition to S4; (3) close relevant valves within a defined maximum response time; (4) remain inside the mechanical capture envelope; (5) remain below defined maximum vertical displacement and velocity; (6) prevent uncontrolled free-float; (7) keep passive support at or above the applicable §6.2 floor. All limits are frozen in ICD-001 and verified at L2/L3. State definition alone is not acceptance.

7.3 REQ-M15 — Manual rescue. A mechanically available two-step rescue mechanism remains operable under the specified power-loss and hydraulic-failure conditions, does not depend on ECU aliveness, requires deliberate two-step activation and creates no unintended free-float. Interface, access location, force, sequence, indication, safe-state outcome and re-enable procedure are configuration-controlled in ICD-001 and verified by physical test.

7.4 REQ-M21 — Lateral-g policy (temporary until SIM-001, HIL and four-corner evidence).

Element	Rule
SPORT→COMFORT inhibit	Immediate when ay exceeds PARAM-M21-AY-HIGH, for automatic and manual requests; driver override cannot defeat it (a request may be queued only)
Re-enable	Only after ay stays below PARAM-M21-AY-LOW for PARAM-M21-HOLD
Filtering and latency	Defined and validated filter and latency budget, biased to inhibit; the former "no time filter" wording is SUPERSEDED
Sensor validity and faults	Validity criteria; defined safe arbitration on invalid, stuck or disagreeing sensors; no single lateral-acceleration sensor can create an unsafe transition
Transients	Noise handling; yaw and roll transients; braking and combined manoeuvres
COMFORT→SPORT	Also conditioned on lateral acceleration, yaw rate, roll rate and macro stroke position limits (parameters TBD under SIM-001/HIL) so seat-band drive and lock engagement under cornering load cannot cause unconstrained motion or lock damage
Roll gradient	PARAM-ROLL-GRADIENT-VL under SIM-001 / HIL
Manual rescue	Separate procedure (§7.3)

7.5 Sensor functional independence. Redundancy means true functional independence, not duplication. Evaluate common-cause failure through: common power supply, ECU, communication bus, software, mounting, environmental exposure, calibration error and vibration. A safety monitor independent of the primary control path shall be evaluated for critical mode arbitration and force-bound enforcement.

7.6 Progressive-stop curve. STR-001 defines a complete force-displacement and energy-absorption curve, not a single peak value: trigger displacement; initial engagement force; progressive force rise; peak force; absorbed energy; residual stroke; temperature sensitivity; manufacturing tolerance; repeatability; post-event inspection criteria. The stops absorb the design energy (§6.4) independently of Lorentz authority.

7.7 REQ-M25 — Horizontal load path. Closed before any integration-ready claim. Define and verify: longitudinal and lateral stiffness; yaw restraint; free play; load sharing between capture frame, links, stops and SPORT locks; braking, lateral, combined vertical/horizontal, crash and propulsion/brake torque-reaction loads; asymmetric four-corner loading. The Lorentz stage is never credited as primary crash or horizontal path.

8. STATE MACHINE

8.1 States.

State	Name	Definition
S0	Safe / Captured / Power-off	No powered release; capture remains
S1	Precharge / Self-test	Capture confirmed; 48 V path checked
S2	COMFORT isolation	Full envelope with mandatory capture
S2S	SPORT locked	All required locks confirmed; stages held

State	Name	Definition
S3	Degraded	Reduced authority; force bounded
S4	Power-loss capture	Passive support; valves fail-closed; quantitative envelope
S5	Impact preparation	Stops prepared; high-rate unlock inhibited
S6	Impact / arrest	Stops active; no magnet-energy dependence
S7	Post-event safe	No automatic re-enable
S8	Maintenance lockout	Physical lockout

8.2 Transitions.

From	To	Condition
S0	S1	Power-on self-test pass
S1	S0	Controlled shutdown
S1	S3	Self-test fail or critical fault
S1	S2	Driver or automatic COMFORT selection
S2	S2S	SPORT request; all required locks report engaged
S2S	S2	REQ-M21 permits; all locks report released
S2	S3	Critical fault, sensor invalidity, thermal derate, lock fault or accumulator pre-charge loss
S2S	S3	Same causes; locks are NOT released by this transition
S3	S2	Fault cleared per ICD-001 rules
S3	S1	Fault cleared and authorised reset
S2 / S2S	S5	Pre-crash sensor class and/or deceleration threshold (OPEN, ICD-001)
S5	S6	Measured impact or stop contact
S5	S2 / S2S	Impact-preparation abort
S6	S7	Event handling complete
S7	S1	Manual clear by authorised personnel only
S1 / S2 / S2S	S8	Physical lockout, only while stationary
S8	S1	Lockout removed by physical key / procedure
Any powered state	S4	Power loss; valves fail-closed; engaged locks spring-applied fail-locked
S4	S1	Power restored and self-test pass (only exit from S4)

S0 is excluded from "Any powered state → S4".

9. SAFETY ANALYSIS

9.1 Screening FMEA. S, O, D use a 1–10 screening scale (inherited, not evidence-based); RPN = S×O×D. SAF-001 replaces these with formal analysis.

Function / failure	Effect	Mitigation	S	O	D	RPN
Carriage jam	Loss of micro authority	Sensors, stops, passive support	9	3	4	108
Link fracture	Local free-motion risk	Capture frame, stops	9	2	5	90
Leveling stuck	Stroke consumed	Level sensor, S3	8	3	3	72
Hose burst	Loss of support/leveling	Pressure monitoring, stops	9	3	3	81
Coil overheat	Derate / shutdown	Temperature model	7	3	3	63
Magnet force-constant loss	Force drop	Temperature limits, derate	7	2	4	56
SPORT lock fails to engage	Remains COMFORT	Status sensors, alert	6	3	2	36
Partial lock (1–2 of 4)	Asymmetric stiffness	Inhibit S2S	8	2	2	32
False COMFORT at high lateral g	Excess roll	Lateral-g / yaw inhibit	7	2	3	42
Horizontal path overload	Capture shear exceeded	REQ-M25, stops	9	2	4	72
Power loss	Active force lost	S4 with §7.2 limits	9	SAF-001	SAF-001	—
Software / OTA fault	Bad command	Bounds, S3/S4	8	3	4	96
Unintended SPORT lock at speed	Handling/comfort conflict	Arbitration inhibit, alert, R-17	8	3	3	72
Accumulator pre-charge loss	Macro response degraded	Pressure sensing, S3, R-18	7	3	3	63
Hydraulic leak / fire	Loss of support, fire risk	Fluid selection, containment, R-19	9	2	4	72
Common-cause sensor fault	False mode or force	Functional independence and independent safety monitor (§7.5)	8	3	3	72
Magnet demagnetisation	Force-constant drop	Temperature limits, derate, R-21	7	2	4	56

9.2 Preliminary HARA (screening; severity class is written Sev-class to avoid confusion with FMEA S).

Hazard	Scenario	Sev-class / E / C	Draft safety goal
Loss of capture	Carriage or link failure	S3 / E3 / C3	Maintain capture under single failure
Uncontrolled vertical motion	Force runaway	S2 / E3 / C2	Bound force; S3 or S4
Asymmetric SPORT lock	Partial lock plus lateral load	S2 / E3 / C2	Inhibit S2S unless complete
Horizontal overload	Brake or crash without defined path	S3 / E2 / C3	Size paths under REQ-M25
Unexpected height drop	Valve open at speed	S2 / E3 / C2	Fail-closed valves
Post-crash automatic re-enable	Uncommanded motion	S2 / E2 / C2	S7, no automatic re-enable
Unintended SPORT lock	False engage on rough road	S2 / E3 / C2	Inhibit, alert, track under R-17
Unlock under high lateral g	Release request in a turn	S2 / E3 / C2	REQ-M21

9.3 Safety process. No ASIL is assigned and no safety case is claimed. SAF-001 evolves into a configuration-controlled safety case: HARA, FMEA, FTA, common-cause and dependent-failure analysis, fault injection, safe-state definition, diagnostic coverage, single-point and latent-fault analysis, verification traceability. Mechanical capture, passive support and progressive stops are independent of E/E and form the primary safety backbone. Cyber and OTA paths able to change force bounds or mode policy require authenticity and integrity controls. Candidate references are in Annex B; no compliance is claimed.

10. ACTUATOR MAP, GAP ENVELOPE AND THERMAL

10.1 Km evidence chain. PARAM-ACT-KM-MIN stays an assumption until validated by measured force-current-power-gap data. The verified ACT-001 map, not nominal Km, is the design input for control and thermal sizing.

Axis	Sweep / points	Record
Air gap	Across the full tolerance envelope, min to max	Force, current, power
Temperature	Cold, nominal, hot	Force, resistance, magnet temperature
Current	RMS region to peak	Force, power, thermal rise
Duty cycle	Continuous and pulsed	Thermal equilibrium
Manufacturing sample	Minimum agreed number of samples	Spread and capability
End-of-life surrogate	If available	Derating note

10.2 Gap tolerance envelope. PARAM-LORENTZ-GAP is a complete tolerance-controlled envelope, not a nominal range. ACT-001 demonstrates force capability and safe control authority across the validated envelope.

Worksheet field	Purpose
Nominal / minimum / maximum gap	Design centre; contact-risk bound; authority bound
Assembly tolerance	Build variation
Structural deflection under load	Operating shift
Thermal growth	Hot/cold shift
Vibration-induced motion	Dynamic envelope
Manufacturing tolerance	Part capability
Sensor error	Control uncertainty
Stop-to-gap relationship	End-of-travel coordination

10.3 Thermal. Continuous thermal design is dominated by copper loss at the RMS operating point; continuous and transient limits are separated. Continuous RMS capability and REQ-M11 stay OPEN until bench evidence closes the model. Demagnetisation (R-21) is evaluated across temperature and duty cycle. SPORT copper demand is near zero (stages held; evidence OPEN).

11. MODE SEQUENCES AND LOCK ARCHITECTURE

11.1 COMFORT→SPORT.

Step	Action	Interlock
1	Arbitration accepts request	No active lateral-g or fault inhibit; §7.4 COMFORT→SPORT conditions met
2	Macro and leveling drive body to seat band	Stroke sensors in range
3	Seat contact detected at required corners	Contact flags true
4	Positive locks command engage	Actuator enable healthy
5	All required locks report engaged	Incomplete set aborts to COMFORT
6	Stages held; state S2S	Driver indication ON

SPORT is never declared while a required lock is disengaged.

11.2 SPORT→COMFORT. Permitted only per §7.4. Locks command release; all report released; macro envelope re-enabled; micro control re-enabled; state S2. COMFORT is never declared while a required lock is engaged.

11.3 Lock architecture (LCK-001).

Element	Function	Status
Seat faces	Upper hardpoint to lower receiver at seat band	Geometry OPEN
Positive lock	Pin, claw or hydraulic clamp preventing separation	Per-corner status required
Actuation	Hydraulic or electromechanical	Duty cycle STUDY
Fail behaviour	Known state with driver alert; never free-float	BASELINE
Crash role	May share load; progressive stops remain primary energy path	BASELINE

12. VERIFICATION FRAMEWORK

12.1 Evidence ladder.

Level	Scope	Status
L0	Architecture and requirements (this document)	BASELINE
L1	SIM-001 numerical run with sensitivity and combined manoeuvres	OPEN
L2	Single-tower bench: force-gap-current, thermal RMS, power loss, lock cycle, stop contact	OPEN
L3	Four-corner / HIL: asymmetry, mode transition, stops, fault injection	OPEN
L4	Optional mule: ISO 2631-1 and handling on donor chassis	OPTION

12.2 SIM-001 specification.

Item	Definition
Model class	Quarter-car / half-car: upper mass, macro k and c, optional micro force; SPORT as high-k constraint
Reference passive	Frozen passive hydro-pneumatic k and c stated on every plot
Roads	ISO 8608 segments, at least one rough and one smooth
Mass states	Empty, occupied, maximum (Annex X mass basis = §6.2 / §6.3 references)
Metrics	aw (0.5–80 Hz), VDV, MSDV (0.1–0.5 Hz); axes, weightings (Wk, Wd; Wf for MSDV) and measurement point (seat surface or floor) stated per CMFSB-TP-001
Outputs	Transmissibility, band RMS, stroke histogram, roll proxy SPORT vs COMFORT, transition time history
SIM-C1	COMFORT lower RMS than frozen passive in at least one agreed band on a rough profile
SIM-C2	SPORT higher roll stiffness (lower roll) than COMFORT on the same lateral step
SIM-C3	Macro within stroke without continuous stop hammering
SIM-C4	No unconstrained free-float interval in transition cases

Results are PRELIMINARY until independent review.

12.3 Power-loss test (REQ-M07). Numeric limits are those frozen in ICD-001; until then results are developmental only.

Step	Action	Record
1	Establish COMFORT at defined payload and temperature	State, payload, temperature
2	Remove commanded active power (controlled cut)	Method, timestamp
3	Measure detection time	ms
4	Measure valve closure time	ms

Step	Action	Record
5	Measure peak vertical displacement and velocity	mm, mm/s
6	Confirm final settle inside capture envelope	Pass/fail vs ICD
7	Confirm no uncontrolled free-float	Pass/fail
8	Confirm passive support adequacy for load class	Pass/fail
9	Repeat from SPORT (S2S)	Full record set
10	Brown-out cases as agreed	Developmental

12.4 Lateral-g interlock test (REQ-M21).

Case	Stimulus	Expected
Threshold inhibit	[ay] steps above PARAM-M21-AY-HIGH	SPORT→COMFORT inhibited
Hysteresis re-enable	[ay] below PARAM-M21-AY-LOW for PARAM-M21-HOLD	Release permitted if other conditions OK
Single sensor fault	One path invalid or stuck	No unsafe release
Redundant disagreement	Paths disagree beyond limit	Safe inhibit or degraded rule
Yaw/roll transient	Combined dynamics profile	No nuisance or unsafe transition
Braking combined	Longitudinal + lateral	Policy-compliant behaviour
Driver override attempt	Override during inhibit	Cannot defeat interlock
COMFORT→SPORT in a turn	Request above the §7.4 conditions	Request held or refused; no free motion

12.5 Progressive-stop inspection (post-event, STR-001).

Check	Criterion	Disposition
Visible deformation	Within STR-001 allowables	Go / no-go for reuse
Residual stroke	Minimum residual retained	Go / no-go
Attachment integrity	No critical fastener or parent failure	Repair / replace
Force-displacement sample	Within curve tolerance if sampled	Accept / investigate
Temperature exposure	Record if extreme	Engineering review
Repeat-event policy	Per STR-001	Replace if required

12.6 Evidence record and reference protocol. Every record retains software version, parameter set, configuration hash, hardware configuration, road/profile ID, payload, mode, passive k and c, time histories, sensor calibration, uncertainty and data-quality flags, and review status. CMFSB-TP-001 defines: vehicle mass; payload; road profile; speed; tyre pressure; passive baseline; COMFORT; SPORT; sensor locations; sampling rate; filtering; metrics; acceptance criteria; uncertainty; configuration hash. Recommended demonstrator: one tower rig showing COMFORT, SPORT lock, power cut to S4, and sensor/actuator fault injection to a safe degraded state.

13. GATES, PHASES, RISKS, WORK PACKAGES AND OPEN ITEMS

13.1 Gates.

Gate	Intent	Entry evidence
G0	Concept controlled	Document issued; legal owner recorded
G1	Interface and simulation ready	ICD-001 draft; SIM assumptions frozen; clash check; Packaging Gate status recorded; REQ-M07 limit placeholders; reviewer appointed
G2	Single-tower evidence	Force, thermal, power-loss, lock bench; ACT-001 map start; REQ-M07 limits frozen
G3	Four-corner / HIL	Mode transitions; stop curve; fault injection; REQ-M25 path
G4	Optional mule	ISO 2631-1 and handling on donor
G5	Licence / transaction readiness	Buyer-accepted package; commercial term sheet; IP filing status; ownership chain

13.2 Phases (planning; no commitment without funding).

Phase	Month	Content
Ph0	0–2	CAD envelope, lock faces, ICD-001 draft; procurement package after ICD baseline
Ph0b	2–6	Supplier lead time (4–8 months planned), COTS surrogate bench, SIM-001 run; 90-day Option deliverables limited to SIM-001 and ICD-001 draft
Ph1	6–10	Single-tower force, thermal, power loss, lock cycle; ACT-001 map start
Ph2	10–15	Four-corner 6-DOF; mode transitions; stop curve; power loss from each mode; REQ-M25 path
Ph3	15–21 (optional)	Donor-chassis ISO 2631-1 and handling

13.3 Risk register (L and I on a 1–5 scale).

ID	Risk	L	I	L×I	Owner
R-01	Core tower mass exceeds limit at PARAM-MASS-CONFIDENCE	4	4	16	Module
R-02	Km not achieved / not demonstrated by measured map	3	5	15	Actuator
R-03	Lorentz thermal or force-gap limits	3	5	15	Actuator
R-04	Macro–micro clash or carriage failure	3	4	12	Mechanical
R-05	Stroke consumed without leveling	3	4	12	Hydraulics
R-06	Regulatory path unclear	3	4	12	Homologation
R-07	Vehicle P90 exceeds host mass ceiling	4	4	16	Mass
R-08	Per-tower crash load versus capture frame, stops and hardpoints (§6.3)	4	5	20	Structures
R-09	Comfort unproven until L1/L2	4	4	16	Simulation / PoC
R-10	EMI or field-exposure cost	2	4	8	EMC
R-11	Hydraulic mass above STUDY value	3	3	9	Hydraulics
R-12	Liability if safety case or claims exceed evidence	2	5	10	Legal / Safety
R-13	SPORT lock reliability or partial engagement	3	4	12	Lock
R-14	Mode-arbitration false positive or negative	3	3	9	Controls
R-15	Horizontal restraint undersized (REQ-M25)	3	5	15	Structures
R-16	Clearance-stack energy underestimated	3	4	12	Structures
R-17	Unintended SPORT lock at speed on rough road	3	4	12	Systems

ID	Risk	L	I	LxI	Owner
R-18	Accumulator pre-charge loss	3	3	9	Hydraulics
R-19	Hydraulic leak or fire	2	5	10	Hydraulics
R-20	Common-cause sensor fault	3	4	12	Systems
R-21	Magnet demagnetisation	2	4	8	ACT-001
R-22	Packaging residual remains negative at engagement	4	4	16	Packaging Gate
R-23	IP filing delayed relative to external disclosure	3	5	15	IP-001

R-08 is the highest screening score.

13.4 Work packages.

Priority	Work package	Deliverable
G1	Freeze CAD envelope, lock faces, leveling, fail-closed valves; Packaging Gate status	3-D package, clash report, residual status
G1	ICD-001 draft (mechanical, electrical, communication, safety, lock) with REQ-M07 placeholders	Buyer-readable ICD set
G1	SIM-001 assumptions and first COMFORT-vs-passive run	Plot pack v0.1 and configuration hash
G2	Single-tower force, thermal, power loss, lock cycle	Bench data package
G2	ACT-001 map start; STR-001 curve; REQ-M25 definition	Maps and structures memo
G2	Formal HARA, FTA, common-cause analysis	SAF-001 work products
G3	Four-corner mode transitions and fault injection	6-DOF records and transition-time records
G5	Licence / IP-transaction package; IP filing status	Commercial pack under counsel (COM-001)

13.5 Document register.

ID	Scope	Status	Gate
DOC-00	This baseline (R5.16)	Issued	G0
MOD-001	Module datasheet (1–3 page buyer summary)	To issue	G0
ICD-001	Interface control	OPEN	G1
SIM-001	Multi-band simulation	OPEN	G1
CFG-001	Configuration management and review disposition	OPEN	G1
HYD-001	Hydraulic macro stage, valves, leveling	OPEN	G1/G2
STR-001	Structure, progressive stops, crash path, REQ-M25	OPEN	G2
LCK-001	Lock system	OPEN	G2
ACT-001	Actuator map, thermal, efficiency	OPEN	G2
ENE-001	Power and current envelope	OPEN	G2
SAF-001	HARA, FMEA, FTA, common-cause, safety case	OPEN	G2
TST-001/002	L2 / L3 test plans	OPEN	G2/G3
BRK-001	Brake and horizontal-load interface	OPEN	G3
REG-001	Regulatory pathway	OPEN	G3
IP-001	IP family, prior art, claim chart, ownership, FTO	OPEN	Counsel
COM-001	Commercial terms (T3)	Counsel-controlled	Option

13.6 Open-item tracker.

ID	Item	Owner class	Gate
O-01	Packaging residual V-L non-negative with all §4.2 lines	Packaging	G1
O-02	REQ-M07 numeric limits in ICD-001	Safety / Systems	G2
O-03	REQ-M15 physical trial under power and hydraulic loss	Safety	G3
O-04	REQ-M21 full policy evidence and PARAM-M21-HOLD	Dynamics	G2
O-05	REQ-M25 horizontal path closure	Structures	G2
O-06	STR-001 curve and design energy event	Structures	G3
O-07	ACT-001 actuator map	Actuator	G2
O-08	ENE-001 total power and current	Electrical	G2
O-09	Formal HARA / FTA	Safety	G2
O-10	Patent filing number and date	IP / Counsel	G5
O-11	Ownership chain and entity assignment records	Legal	G5
O-12	Independent reviewer appointment (PARAM-REVIEWER-DEADLINE)	Programme	G1
O-13	REQ-M11 temperature range	Thermal	G2
O-14	REQ-M12 EMC / cabin field	EMC	G3
O-15	PARAM-MASS-CONFIDENCE and aggregate P90 method	Mass	G1
O-16	Mode-time allocation (lock time, inhibit exclusion)	Controls / Lock	G2

14. HOST VARIANTS (PATH B)

Host variants are application-layer references. Core module IP does not depend on V8, V12, 4×4, 6×6 or CFRP, and Path A does not require any host figure. LOCKED means engagement decision, not verification, not packaging closure; the Packaging Gate (§4) applies. Engine entries are class descriptors, not brands.

14.1 Geometry and powertrain.

Parameter	V-L Urban Light	V-H Military / Off-road	Status
Length	5,000 mm	5,500 mm	LOCKED
Width (mirrors excluded)	2,000 mm	2,200 mm	LOCKED
Height (SPORT / seated)	1,700 mm	2,000 mm	LOCKED
Wheelbase	3,000 mm	3,300 mm (front axle to rear-tandem midpoint)	LOCKED
Ground clearance (SPORT)	200 mm	300 mm	LOCKED
Drive	4×4	6×6 (front axle + dual rear tandem)	LOCKED
Design occupants	4	6	LOCKED
Passenger cell	Full CFRP structural cell	Full CFRP structural cell	BASELINE

Parameter	V-L Urban Light	V-H Military / Off-road	Status
Roll structure	FIA-referenced CFRP/hybrid (no homologation)	Same	Reference only
GVW ceiling	PARAM-GVW-VL	PARAM-GVW-VH	LOCKED
Powertrain	Hybrid ICE + e-drive	Hybrid ICE + e-drive	LOCKED
ICE class	V8, 4.0 L twin-turbo class	V12, 6.0 L twin-turbo class	LOCKED class
ICE power	≈430–450 kW class	≈460–500 kW class	TARGET
Electric traction	≈150–180 kW	≈200–250 kW	TARGET
Battery usable / fuel tank	20 kWh / 80 L	40 kWh / 150 L	LOCKED
Range	1,000 km	700–800 km	STUDY (recompute road load at 90/120 km/h in ENE-001; V-L ≈8 L/100 km is optimistic)

14.2 Carbon cell and roll structure. Conventional CFRP layup and bonded construction is baseline until continuous-fibre printing is evidenced. Cell mass budget ≤280 kg (V-L) and ≤420 kg (V-H), TARGET. Roll-structure mass budget ≤45 kg and ≤70 kg. Roll-structure philosophy may reference FIA Appendix J concepts subject to verification of article numbers and edition; load cases are OPEN until edition freeze.

14.3 Class mapping and integration. V-L maps to Class S; V-H to Class M/L. A 4×4 pilot is generally the shorter first-contract path; 6×6 is a funded second SKU with higher axle, steering and packaging risk, optional on V-H only if axle mass is traded from reserve or protection. Hardpoints match host chassis stiffness; brake and propulsion torque reactions shall not overload horizontal links.

14.4 Closed mass budgets (sole valid host budget; STUDY).

WBS line	V-L kg	V-H kg
CFRP passenger cell	280	420
Roll members	45	70
Closures, glass, panels / reinforced exterior	230	400
Interior, HVAC, cabin LV	200	330
Lower chassis	340	720
ICE and ancillaries	275	390
Hybrid transmission / coupling	160	190
E-motors and inverters	130	180
Battery (usable energy per §14.1)	145	260
HV/LV and cooling	95	160
Axles, steering / three-axle driveline	260	580
Wheels, tyres, brakes	210	480
Isolation, four corners (Class S / M-L)	280	380
Optional light protection package	—	200
Fuel tank and reference fuel	75	140
Exhaust and fluids	70	100
Programme reserve	70	150
Curb mass, no occupants	≈2,865	≈5,150
Occupants (PARAM-OCC-MASS × 4 / × 6)	400	600
Cargo / mission payload	100	400
GVW design case	≈3,365	≈6,150
Margin to ceiling	≈635	≈850

V-H includes the optional 200 kg protection package; if omitted it moves to reserve. REQ-V01/V02: GVW claims stay within the ceilings including the occupant policy. REQ-V03: both variants use a CFRP cell within these budgets. REQ-V04: FIA-referenced roll structure without homologation claim. REQ-V05: external materials name V-L and/or V-H whenever mass is quoted.

15. SUPPLIER AND ICD OUTLINE

15.1 Supplier data requests (no supplier is committed by this document; catalogue values are not programme evidence).

#	Domain	Request
1	Lorentz	Force-current-gap-temperature maps, or commitment to support ACT-001; continuous and peak force; gap tolerance; thermal derating curves
2	All	Mass P50/P90 with process-capability statement
3	Hydraulics	Cylinder force at pressure; seal life; fail-closed valve response and leakage class; duty, contamination and service intervals; hold-option capability
4	Locks	Engage/release time under load; status sensing; shear capacity; contamination tolerance
5	Stops	Force-displacement curve; energy rating; temperature sensitivity; inspection interval
6	Sensors	Stroke, force, pressure, temperature accuracy and fault behaviour; independence architecture
7	Electronics	48 V drive capability; current limit; diagnostic coverage; EMC data or test plan
8	Commercial	Lead time; change control; IP-clause willingness; quality system and traceability level

15.2 ICD-001 minimum chapters. Hardpoint geometry and loads; hydraulic ports and fluid; electrical power and signals (48 V class with full current envelope); mode and interlock logic; progressive-stop curve reference; REQ-M07 limits; thermal limits; installation torque and sequence; maintenance and rescue release; service access; EMC and grounding; configuration identity.

16. TRANSACTION READINESS (PATHS A, B, C)

16.1 Transaction paths.

Path	Object	Typical inclusions	Not included unless separately agreed
A — Licence / supply	Right to use the CMFSB module architecture in a named field	DOC-00 and agreed controlled subset; ICD, datasheet, safety behaviour, dual-mode rules; agreed engineering support hours; updates within a defined revision window; Option milestones	Assignment of all IP; certified vehicle or type approval; unlimited R&D; production tooling; guaranteed comfort percentage; guaranteed regulatory approval
B — Vehicle programme	Module plus host variant V-L and/or V-H	Host packs (§14)	Vehicle certification packs, series production

Path	Object	Typical inclusions	Not included unless separately agreed
C — IP assignment	Assignment of the module IP family	Bilateral terms subject to counsel, filing status and evidence gates	Anything not in the assignment schedule

Default Path A Option: non-exclusive evaluation in a named field; limited exclusivity as a paid add-on. Contracting party is Amirbahador Atai (individual) until formal assignment to a legal entity; the entity path (owner, target date) is recorded under IP-001 and COM-001 before any Option is signed, and liability allocation is defined in COM-001.

16.2 Readiness boundary.

Use case	Suitable on this document	Condition
NDA and T1 diligence	Yes	Under NDA
Paid Option negotiation	Yes	Bilateral term sheet in COM-001
Module licence term sheet	Conditional	ICD-001 draft plus SIM-001 or a committed bench plan
Large upfront assignment or exclusive sale	Not on this document alone	IP filing status, ownership chain, counsel FTO review, evidence gates
Production or type-approval claim	No	Non-claims §1.2
Advertising measured ride comfort	No	Until L1/L2 configuration-managed results

16.3 Buyer diligence matrix.

Buyer question	Current answer	Evidence / gate
Is the architecture defined and safe by construction?	Mechanical capture, passive support and independent stops are architectural invariants (§2.3)	SAF-001, G2/G3
Does it work?	Not demonstrated; TRL 2–3	SIM-001 (L1), bench (L2)
Is the actuator real?	Km is an assumption	ACT-001 map
Does it fit?	V-L stack negative; V-H +10 mm before missing lines	Packaging Gate, O-01
What does it weigh?	Core tower 48 kg P50; 67 kg line-sum P90 vs limit	O-15, supplier data
Who owns it?	Amirbahador Atai (individual)	IP-001 ownership chain, O-11
Is it protected?	Filing status per Annex T3-A.4	O-10
What is the safety status?	Screening only; no ASIL	SAF-001

16.4 Buyer information set (minimum for a serious discussion under NDA).

Element	Source	Tier
This baseline	DOC-00 R5.16	T1
Module datasheet extract	MOD-001	T1
Safety behaviour description (S4, locks, stops)	§2, §7, §8	T1
ICD draft or outline	ICD-001	T1/T2
SIM-001 assumptions list	SIM-001	T1/T2
Open-item list	§13.6	T1
IP overview without claim charts	§16.5	T1
IP schedule, claim charts, prior-art mapping	Annex T3-A	T3
Commercial term sheet	COM-001	T3

16.5 IP statement for T1 distribution. Multi-family IP protection is in preparation under counsel. No statement of patent novelty, enforceability or freedom-to-operate is made. IP protection proceeds in parallel with engineering and does not wait for prototype completion. No T1 document is released to a third party before a provisional filing or written counsel clearance.

16.6 Conditions precedent to any assignment or exclusive licence (checklist).

#	Condition	Evidence
1	Provisional (or later) filing(s) made and recorded	Application number, office, date
2	Inventor declarations and chronology complete	IP-001
3	Assignments and contractor / supplier IP clauses documented	Ownership chain file
4	Confidentiality and disclosure history documented	NDA register
5	Prior-art search and claim chart reviewed by counsel	Annex T3-A
6	FTO review by qualified counsel	Counsel opinion
7	Independent technical reviewer appointed and findings dispositioned	CFG-001
8	Entity assignment path defined	IP-001 / COM-001

17. GOVERNANCE AND SIGN-OFF

17.1 Configuration control. One controlled source per numerical parameter (Section 3); every critical parameter has a unique ID; historical values are retained only as SUPERSEDED. CFG-001 minimum content for each controlled change:

Item	Required content
Document revision	ID, date, supersedes
Parameter change	Old value, new value, reason
SUPERSEDED mark	Prior value retained for traceability
OPEN-item closure	Evidence reference and gate
Review finding	Disposition and owner
Evidence configuration hash	Software, parameters, hardware serial class
Disclosure tier	T0–T3

DOC-00 is the narrative baseline; design freeze occurs in ICD-001 and part drawings, not in DOC-00 alone. All TARGET and STUDY values that become design inputs are configuration-managed.

17.2 Independent reviewer challenge list (before G1 credit).

#	Challenge topic
1	Capture claims versus free-levitation misunderstanding
2	Mass P50 versus confidence-level consistency
3	Km assumption exposure in external materials
4	Static floor rule and mass-basis consistency

#	Challenge topic
5	Packaging residual versus LOCKED language
6	REQ-M07 measurability (limits frozen?)
7	REQ-M15 operability under dual failure
8	REQ-M21 single-sensor and disagreement behaviour
9	REQ-M25 openness versus any integration-ready language
10	Progressive-stop curve completeness and design energy event
11	IP ownership chain completeness
12	Configuration discipline and one-value-one-place
13	State-machine completeness (S3 entry, S4 exit)

17.3 Release and sign-off.

Role	Name / status	Date / target
Legal Owner / Author	Amirbahador Atai	2026-10-06
Independent Technical Reviewer	TBD	PARAM-REVIEWER-DEADLINE
Approval Authority	TBD after independent review	After G1 evidence
Documentation authority	ATFSV Project Engineering Documentation Authority	2026-10-06

Status: CONTROLLED / Contract Diligence Baseline. Not approved for road use, production or certification credit. All TARGET, STUDY, ASSUMPTION, DERIVED, LOCKED and OPTION values require formal verification before design release. Priority before serious OEM or investor engagement: (1) ACT-001, STR-001 and SIM-001 evidence; (2) SAF-001 including quantitative power-loss limits, sensor independence and horizontal load path; (3) IP-001 filing, prior art, claim chart and ownership chain.

ANNEX T3-A — IP SCHEDULE (T3 — COUNSEL CONTROLLED — REMOVE FROM ALL T1/T2 DISTRIBUTION)

T3-A.1 Proposed patent families (for counsel review).

Family	Scope	Treatment
A	Core mechanically captured macro–micro isolation architecture	Patent candidate
B	Hybrid hydraulic macro + high-bandwidth micro actuator architecture	Patent candidate
C	Positive SPORT lock and mechanical capture geometry	Patent candidate
D	Safety state machine; S4 passive-support with quantitative envelope; transition constraints	Patent / trade-secret review
E	Dynamic mode arbitration and control; REQ-M21 interlock logic	Patent / software-protection review
F	Progressive-stop and independent crash path with full force-displacement absorption	Patent candidate
G	Modular host-class integration interface (four-tower ICD, load sharing)	Patent candidate
Trade secrets	Actuator tuning, calibration maps, control parameters, manufacturing processes, supplier-specific implementation	Trade-secret candidate

T3-A.2 Invention-disclosure candidates (subject to counsel): (1) mechanically captured macro–micro isolation architecture; (2) passive mechanical support combined with high-bandwidth Lorentz micro-actuation; (3) dual COMFORT/SPORT operating architecture; (4) positive mechanical SPORT lock architecture; (5) safety state machine and transition constraints; (6) S4 passive-support architecture with quantitative envelope; (7) progressive-stop and independent crash-load path with full force-displacement curve; (8) horizontal-load restraint architecture; (9) integration interface across multiple host classes; (10) coordinated control method combining vehicle dynamics, isolation state and mechanical locking.

T3-A.3 Four-layer IP structure and differentiation matrix.

Layer	Content	Purpose
1 Core architecture	Capture + macro isolation + micro actuation + stops	Survives change of actuator technology
2 Dynamic behaviour	COMFORT/SPORT, transition logic, lateral-g policy	Product differentiator
3 Safety architecture	S4, independent crash path, fail-closed, positive lock	OEM safety-case relevance
4 Integration platform	Four-tower interface, loads, modularity	Licensing across host classes

Feature	Prior art	What is different	Patent candidate	Trade secret
Mechanical capture	TBD by counsel	Always kinematically constrained; no free levitation	Yes	No
Macro–micro isolation	TBD by counsel	Hybrid hydraulic + high-bandwidth micro stage inside mechanical envelope	Yes	Tuning, maps
SPORT lock	TBD by counsel	Positive mechanical lock for handling stiffness	Yes	No
S4 power loss	TBD by counsel	Fail-closed + passive support + quantitative envelope	Yes	No
REQ-M21 logic	TBD by counsel	Lateral-g interlock with full sensor, fault and transient policy	Yes	Parameters
Progressive stop	TBD by counsel	Independent of magnetic closed loop; full curve	Yes	No
Module interface	TBD by counsel	Host-class adaptable four-tower ICD	Yes	No

T3-A.4 Filing and ownership record (to be completed and dated by counsel).

Item	Record
Provisional application(s)	None filed as of 2026-10-05 per project record. Update: No. [], Office [], Date []
Inventor(s)	Amirbahador Atai [confirm complete list and chronology]
Current owner	Amirbahador Atai (individual)
Assignment to entity	[Entity, date, instrument — TBD]
Contractor / supplier IP clauses	[TBD]
Confidentiality and disclosure history	[NDA register — TBD]
Jurisdiction strategy	[TBD by counsel]
FTO opinion	[TBD by counsel]

IP-001 content: invention disclosure, inventor chronology, prior-art search, patent landscape, claim chart, potentially conflicting families, jurisdiction strategy, ownership chain, assignment records, confidentiality history, FTO review.

T3-A.5 Term-sheet topic schedule (topics only; values are bilateral and live in COM-001).

Field of use · territory · exclusivity and its fee · Option period and exercise mechanics · milestone and royalty structure · improvements and grant-back · ownership and title warranties · IP representations and indemnity · data-room and escrow arrangements · supplier flow-down · audit rights · export-control compliance · confidentiality and publication · termination and reversion · governing law and dispute resolution.

ANNEX B — REFERENCES AND GLOSSARY

Reference	Use	Status
ISO 26262	Functional-safety process	Candidate; compliance not claimed
ISO 21448	Safety of the intended functionality	Candidate
ISO 2631-1	Whole-body vibration evaluation	Candidate
ISO 8608	Road-surface profiles for simulation	Candidate
ISO 4413	Hydraulic fluid power safety (REQ-M24)	Reference
ISO 13849-1	Lock and valve safety functions if used	Candidate
UN R100 / ISO 6469-3	HV safety if applicable	Candidate
UN R94 / R95 / R14 / R16	Crash and restraint matters if applicable	Candidate
EN 1789 / EN 1865	Ambulance planning references (Application A only; no clinical claim)	Candidate
FIA Appendix J	Roll-structure philosophy only	Reference

Glossary: CMFSB — Captured Mechanical Floating Safety Body. S4 — power-loss passive-support state. COMFORT / SPORT — dual running-gear modes (S2 / S2S). Application A — ambulance and clinical-transport planning context. Path A / B / C — licence, vehicle programme, IP assignment. VDV — vibration dose value. MSDV — motion sickness dose value. P50 / P90 — median and 90th-percentile planning values.

END OF ATFSV-DOC-00-R5.16

AT-FSV / ATAI

CMFSB ISOLATION MODULE — DOCUMENT 00

MASTER ENGINEERING BASELINE | ATFSV-MOD-001 | Captured Mechanical Floating Safety Body

Revision 5.16 — Single-Layer Controlled Baseline for Licence and IP-Transaction Negotiation

Issue date: 2026-10-06

Legal Owner / Author: Amirbahador Atai

DOCUMENT CONTROL

Field	Controlled value
Document ID	ATFSV-DOC-00-R5.16
Supersedes	All earlier ATFSV-DOC-00 issues and variants R5.11–R5.15, including the 20-page R5.12, the 59-page R5.12-A, the 64-page package, the 55-page integrated file, the 18-page R5.13, R5.13-AMENDMENT, R5.14 and R5.15
Document class	Module-Primary External Baseline
Classification	T1 — Confidential / NDA controlled. Annex T3-A is T3 and shall be removed from any T1/T2 distribution
Legal Owner / Author	Amirbahador Atai (individual; assignment to a legal entity is a separate controlled act)
Documentation authority	ATFSV Project Engineering Documentation Authority (documentation role only; not a legal entity)
Status	Contract Diligence Baseline; not certified; TRL 2–3
Independent Technical Reviewer	TBD — appointed no later than the earlier of 2026-12-05 and G1 closure, and before any external safety-performance credit
Approval authority	TBD after independent review
Next revision trigger	ICD-001 draft · SIM-001 run · STR-001 stop curve · ACT-001 actuator map · ENE-001 envelope · IP filing-status update · reviewer disposition

Controlled limitation. This document supports T1 technical diligence, supplier evaluation, licence negotiation and IP-transaction negotiation. It is not a performance certificate, type approval, safety case, measured ride-comfort claim, production commitment, warranty, patent grant, price list or regulatory approval.

CONTENTS

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- 8 State Machine
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- 11 Mode Sequences and Lock Architecture
- 12 Verification Framework
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- 14 Host Variants (Path B)
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- 16 Transaction Readiness (Paths A, B, C)
- 17 Governance and Sign-Off
- Annex T3-A IP Schedule (counsel controlled)
- Annex B References and Glossary

0. REVISION AND SUPERSESSION

- 0.1 Identity rule. One document ID identifies exactly one content. All earlier files that carry IDs R5.11–R5.15 are WITHDRAWN and are not active design inputs. Only values in Section 3 and requirement text in Section 7 are active.
- 0.2 Corrective items and engineering status. "Applied" means corrective text is in this document. It does not mean engineering closure.

ID	Topic	Applied in	Engineering status
C-01	LOCKED geometry is not packaging closure	\$4	OPEN — V-L residual –120 mm
C-02	6.3 kW is Lorentz copper loss only	\$5	OPEN — ENE-001
C-03	48 V current budget	\$5	OPEN — ENE-001 / ACT-001
C-04	REQ-M07 quantitative criteria	\$7.2	OPEN — limits not frozen
C-05	REQ-M15 under power and hydraulic loss	\$7.3	OPEN — physical trial
C-06	REQ-M21 full sensor/fault/transient policy	\$7.4	OPEN — SIM/HIL evidence
C-07	Sensor functional independence	\$7.5	OPEN — SAF-001
C-08	Km evidence chain	\$10.1	OPEN — ACT-001 map
C-09	Gap tolerance envelope	\$10.2	OPEN — ACT-001
C-10	Progressive-stop curve	\$7.6	OPEN — STR-001
C-11	REQ-M25 horizontal path	\$7.7	OPEN — STR-001 / ICD-001
C-12	Multi-family IP, ownership, disclosure	\$16, Annex T3-A	OPEN — filing status below
C-13	Tower mass statistical basis	\$6.1	OPEN — supplier data
C-14	Static floor rule, rounding and traceability	\$6.2	Applied; inputs STUDY
C-15	Stack-up arithmetic and missing lines	\$4	OPEN
C-16	State machine gaps (S2/S2S→S3)	\$8	Applied; verification OPEN
C-17	"No time filter" removed from REQ-M21	\$7.4	Applied; parameters TBD
C-18	Mode-time allocation	\$7.1, \$11	Applied; values TBD

1. PURPOSE, NON-CLAIMS, VALUE AND DISCLOSURE TIERS

- 1.1 Purpose. This baseline defines ATFSV-MOD-001: a mechanically captured, dual-mode isolation module for specialty, defence-utility, ambulance and limited-series vehicles. It fixes architecture, requirements, evidence plan, risks, interfaces, host references and the transaction boundary.

1.2 Non-claims register (single location).

#	Non-claim
1	No production readiness or series manufacturing capability
2	No regulatory approval, type approval or certification credit
3	No certified ASIL or equivalent safety integrity level
4	No measured ride-comfort percentage or guaranteed comfort improvement until L1/L2 evidence
5	No crashworthiness or survivability demonstration
6	No guaranteed vehicle mass compliance (4,000 kg / 7,000 kg ceilings are engagement targets)
7	No final supplier mass, cost or lead time
8	No closed mass or energy budget suitable for production release
9	No continuous public-road 250 km/h capability and no continuous RMS force capability until thermal bench data exist
10	No clinical outcome claim for ambulance use
11	No patent grant, novelty, enforceability or freedom-to-operate conclusion before counsel-led analysis
12	No FIA homologation or series eligibility from FIA-referenced design philosophy
13	No supplier or OEM endorsement
14	No occupant-escape or detachable-capsule system is part of this module baseline
15	No warranty, price list or offer of exclusive sale by this document

Contradictory external presentations shall be withdrawn. Sales language may describe architecture and modes and may not invent numbers.

1.3 Value statement. Occupant compartments in demanding vehicles receive road vibration that affects comfort, task performance and, where relevant, clinical stability. Soft isolation alone reduces handling stiffness in rapid manoeuvres. CMFSB combines a COMFORT isolation envelope with a SPORT mechanical lock, keeps the upper body mechanically captured at all times, and settles on passive support on power loss. Measurable advantage is not claimed until L1/L2 evidence exists.

1.4 Planning KPIs (method definitions only).

KPI	Metric definition	Method	Status
Ride isolation	aw (0.5–80 Hz), VDv, MSDV (0.1–0.5 Hz) per ISO 2631-1	SIM-001, then bench	OPEN until L1
Handling stiffness	Roll angle and lateral response, SPORT vs COMFORT, same step	SIM-001, mule	OPEN until L1
Mode time	PARAM-MODE-TIME, §7.1 definition	Bench / HIL	TARGET
Power loss	S4 with quantitative envelope (§7.2)	Power-cut test	BASELINE
Module mass	PARAM-TOWER-MASS-LIMIT at PARAM-MASS-CONFIDENCE	Weighing	TARGET

1.5 Disclosure tiers.

Tier	Content	Access
T0	Public non-enabling concept / teaser	Public
T1	This baseline: architecture, requirements, risk, verification planning	NDA
T2	Models, detailed ICD, test data, configuration-controlled design artefacts	NDA + need-to-know
T3	Claim charts, prior-art mapping, IP schedule, negotiation strategy, commercial terms (COM-001)	NDA + counsel + restricted list

Disclosure is minimised to the lowest tier required. Detailed actuator geometry, control parameters and claim-sensitive implementation stay in T2/T3.

1.6 Status vocabulary.

Label	Meaning
BASELINE	Architectural decision for this revision; verification still required
TARGET	Numeric design goal; not verified
STUDY	Indicative planning figure; may change
ASSUMPTION	Input without programme-owned evidence
DERIVED	Calculated from stated inputs; inherits their uncertainty
LOCKED	Programme engagement decision; not test verification and not packaging closure
OPEN / TBD	Unresolved at this revision
OPTION	Not committed
SUPERSEDED / HISTORICAL	Retained for traceability only

2. ARCHITECTURE, SAFETY INVARIANTS AND MODES

2.1 Architecture. The upper body is kinematically constrained at all times (no free levitation). Four corner towers each combine: a hydraulic macro stage (PARAM-M02-MACRO-STROKE) with leveling and lock seats; a Lorentz micro stage (PARAM-M03-MICRO-STROKE) within the PARAM-LORENTZ-GAP envelope; progressive mechanical stops; and an optional SPORT positive lock. A shared ECU performs mode arbitration, the REQ-M21 interlock and power-loss detection. Layer stack, top to bottom: upper occupant/safety body → micro stage → macro stage → lower chassis. Architectural signature: the body is always mechanically captured; active isolation operates inside a bounded mechanical safety envelope; comfort control, handling stiffness and crash containment are coordinated but independently bounded. This signature stays valid if the Lorentz micro stage is replaced by another high-bandwidth actuator.

2.2 Independent load and force paths.

Path	Carrier
Normal vertical load	Passive / compliant mechanical-hydraulic support
Comfort dynamic force	Lorentz micro stage (force-bounded)
Large displacement and leveling	Hydraulic macro stage
Crash / abnormal load	Mechanical capture + progressive stops (independent of Lorentz closed loop)
Horizontal load	Capture frame, links, horizontal stops, SPORT lock faces (§7.7)
Power loss	S4 passive support with quantitative envelope (§7.2)

2.3 Safety invariants.

Invariant	Rule
Mechanical capture	No operating state permits loss of geometric capture
Static support	Vehicle weight on passive or fail-safe compliant paths, never solely magnetic
Power loss	S4 passive support; valves fail-closed; software aliveness not required

Invariant	Rule
Crash path	Progressive stops and structure independent of closed-loop magnetic force
Mode transition	No transition enters unconstrained free-float
SPORT lock fault	Lock failure shall not defeat capture or progressive stops
Horizontal restraint	Dedicated paths and/or SPORT lock; REQ-M25 closes before any integration-ready claim
Override limit	Driver override cannot defeat the REQ-M21 interlock
Manual rescue	Two-step mechanism operable under power loss and hydraulic failure; no unintended free-float

2.4 Modes.

Mode	State	Mechanical condition	Objective
COMFORT	S2	Locks released; macro and micro active; leveling active	Soft ride isolation
SPORT	S2S	Upper body seated; all required locks confirmed; stages held	Handling stiffness

Selection inputs: driver command (highest priority), radar/surface proposal, GPS/map proposal. Default after self-test is COMFORT unless ICD-001 states otherwise. Hold function (optional): rate-limited hydraulic orifice or friction device at the macro stage; not a primary crash path; velocity and force limits are frozen in ICD-001; if not fitted, REQ-M16 is N/A.

2.5 Dynamics screening (STUDY). With PARAM-REF-MASS-OCC and four corners, corner share is 662.5 kg; with PARAM-K-EFF the natural frequency is PARAM-FN-STUDY. A payload change of 2.4 kN gives about 60 mm at PARAM-K-EFF, so active leveling (REQ-M22) is required. Static sag is carried by gas-spring preload and leveling, not by the dynamic stroke. Carriage first structural mode shall be at least three times the micro control bandwidth.

3. MASTER PARAMETER REGISTER (one value, one place)

Other sections cite the parameter ID and do not restate the value.

Parameter ID	Value	Status
PARAM-M02-MACRO-STROKE	±50 mm body-relative (100 mm range)	TARGET
PARAM-M03-MICRO-STROKE	±7.5 mm	TARGET
PARAM-LORENTZ-GAP	2–5 mm tolerance-controlled envelope (15–30 mm applies to PM+EM OPTION only)	TARGET
PARAM-FORCE-PEAK	±2.5 kN per tower	STUDY
PARAM-FORCE-RMS	≈0.7 kN per tower	STUDY
PARAM-ACT-KM-MIN	≥63 N/√W	ASSUMPTION until ACT-001
PARAM-COPPER-MEAN	≈123 W per tower; ≈0.5 kW four towers (limit ≤0.5 kW) at PARAM-FORCE-RMS	DERIVED / TARGET
PARAM-COPPER-PEAK	≈1.57 kW per tower; ≈6.3 kW four towers; copper loss only	DERIVED
PARAM-SUPPLY-NOM	48 V class	STUDY
PARAM-BW-CONTROL	≥20 Hz; carriage first mode ≥3×	TARGET
PARAM-K-EFF	≈40 kN/m about the leveled operating point	STUDY
PARAM-FN-STUDY	≈1.23 Hz	DERIVED / STUDY
PARAM-REF-MASS-OCC	≈2,650 kg occupied upper body	STUDY
PARAM-REF-MASS-MAX	≈3,150 kg maximum-study upper body	STUDY
PARAM-TOWER-MASS-P50	48 kg	STUDY
PARAM-TOWER-MASS-LIMIT	≤55 kg	TARGET at PARAM-MASS-CONFIDENCE
PARAM-MASS-CONFIDENCE	TBD in CFG-001 before G1 (candidate P90)	OPEN
PARAM-TOWER-MASS-P90-LINESUM	67 kg (conservative line-sum of element P90 values)	STUDY
PARAM-STATIC-RULE	Floor = round-up to 0.5 kN of 1.25 × maximum static corner load	BASELINE
PARAM-STR-S-STATIC-FLOOR	≥5.0 kN per corner	TARGET
PARAM-STR-M-STATIC-FLOOR	≥7.0 kN per corner	TARGET
PARAM-STR-L-STATIC-FLOOR	≥10.0 kN per corner	TARGET (9.7 kN is SUPERSEDED)
PARAM-M21-AY-HIGH	0.35 g	Temporary TARGET
PARAM-M21-AY-LOW	0.30 g	Temporary TARGET
PARAM-M21-HOLD	TBD (SIM-001 / HIL)	OPEN
PARAM-ROLL-GRADIENT-VL	≤6.0 deg/g at V-L design payload	TARGET
PARAM-MODE-TIME	≤3 s, accepted request to declared state, excluding REQ-M21 inhibit time	TARGET
PARAM-LIFE	≥10,000 h equivalent road duty	TARGET
PARAM-OCC-MASS	100 kg per occupant	Policy
PARAM-GVW-VL	4,000 kg ceiling	LOCKED (engagement)
PARAM-GVW-VH	7,000 kg ceiling	LOCKED (engagement)
PARAM-REVIEWER-DEADLINE	Earlier of 2026-12-05 and G1 closure	Governance

4. PACKAGING FEASIBILITY GATE

4.1 Rule. LOCKED vehicle envelope values (overall height, ground clearance and related host geometry in §14) are engagement targets and are not packaging closure. Until H-point, cabin height, tower integration, macro travel, free play, stop stroke, micro-stage clearance and chassis integration are configuration-controlled, geometry remains under this Gate. No production or integration claim is made until the complete vertical and horizontal stack-up closes with a non-negative verified margin. Do not force a non-negative residual by reducing cabin height below H-point requirements.

4.2 Vertical stack-up (SPORT / seated).

Element	V-L (mm)	V-H (mm)	Status
Overall height	1,700	2,000	LOCKED (engagement)
Ground clearance	200	300	LOCKED (engagement)
Available for structure	1,500	1,700	DERIVED
Chassis / axle structure	280	320	STUDY
Tower structure and mounts	120	140	STUDY
Macro travel (100) + free play (40)	140	140	TARGET + STUDY

Element	V-L (mm)	V-H (mm)	Status
Cabin interior height from H-point	≥1,000	≥1,000	TARGET
Roof / exterior shell	80	90	STUDY
Allocated sum	≥1,620	≥1,690	DERIVED
Residual margin (before lines below)	≤ -120	≤ +10	OPEN — P0
Stop stroke (STR-001)	not yet in stack	not yet in stack	OPEN
Micro-stage clearance (PARAM-M03-MICRO-STROKE)	not yet in stack	not yet in stack	OPEN
Tolerance, thermal and static-sag allowance	not yet in stack	not yet in stack	OPEN

V-H residual is exactly +10 mm with the stated numbers; any stop stroke above 10 mm makes it negative unless the stroke is shown to lie inside the tower allowance. Closure options: integrate tower structure inside chassis depth; raise overall height; reduce ground clearance; package free play in the underfloor.

5. POWER AND CURRENT ENVELOPE

5.1 Rule. PARAM-COPPER-PEAK is Lorentz copper loss at simultaneous peak force on four towers. It is not module electrical demand. Electrical input power is copper loss plus mechanical power plus driver/converter losses. Total system power shall include: Lorentz actuators; hydraulic pump; valve actuation; lock actuation; ECU and sensors; transient and inrush; driver/converter losses; safety and thermal derating margins.

5.2 Current. PARAM-COPPER-PEAK divided by 48 V gives ≈131 A, an indicative lower bound for bus current, not a design current. ENE-001 and ACT-001 shall define: continuous, RMS and peak current; peak duration; simultaneous-tower demand; driver efficiency; bus voltage tolerance; cable and connector losses; fuse/protection strategy; thermal derating; transient and regenerative conditions.

5.3 Power budget.

Consumer	Mean (W)	Peak (W)	Closure
Lorentz, four towers	≤500	≈6,300 (copper only)	ACT-001
Hydraulic pump	OPEN	OPEN	HYD-001
Valves	OPEN	OPEN	HYD-001
Lock actuators	OPEN	OPEN	LCK-001
ECU and sensors	OPEN	OPEN	ICD-001
Total system	OPEN	OPEN	ENE-001

6. MASS, STATIC FLOORS AND LOAD CASES

6.1 Module mass and statistical basis. PARAM-TOWER-MASS-LIMIT applies to the core tower only. If it is a production design limit, compliance shall be shown at PARAM-MASS-CONFIDENCE, not at P50. P50 is not "agreed confidence". PARAM-TOWER-MASS-P90-LINESUM is a line-sum of element P90 values and is a conservative upper bound; aggregate P90 shall be computed from element distributions (RSS or Monte-Carlo) when supplier data exist. Risk R-01 tracks this.

Element	P50 kg	P90 kg	Scope
Local structure per tower	18	25	Core
Lorentz and local carriage	22	30	Core
Sensors and local electronics	8	12	Core
Core tower subtotal	48	67	Limit TARGET; P90 under R-01
Cylinder and accumulator share	15	22	Corner
Valves, hoses, seals	5	8	Corner
Progressive stops	8	12	Corner
SPORT lock share	6	10	Corner
Harness, dampers, bellows	4	6	Corner
Pump and reservoir share (¼)	8	12	Shared
Full corner, Class L	≈94	≈137	STUDY
Four corners, Class L	≈376	≈548	Vehicle WBS isolation line
Per-class corner P50 (planning)	S ≈70 · M ≈85 · L ≈95		M interpolated until supplier data

6.2 Static support floor. Rule: PARAM-STATIC-RULE. Mass basis for each class is stated so every floor is traceable.

Class	Mass basis	Maximum static per corner	1.25 × max	Floor (0.5 kN round-up)
S (V-L)	Occupied corner share 310–350 kg gives 3.0–3.4 kN; planning envelope with payload and margin	≈4.0 kN (STUDY)	5.0	≥5.0 kN
M (intermediate)	Corner share 450–550 kg gives 4.4–5.4 kN	≈5.5 kN (STUDY)	6.88	≥7.0 kN
L (V-H / generic)	PARAM-REF-MASS-MAX ÷ 4 × 9.81	≈7.7 kN (7.73)	9.66	≥10.0 kN

Class L inputs come from the generic maximum-study reference mass, which is more conservative than the V-H budget in §14.4 (upper body about 2,220–2,420 kg, ≈5.4–5.9 kN per corner). The generic reference is retained deliberately and is not derived from §14.

6.3 Reference load cases. Class L is shown on both mass bases; crash cases shall state which basis is the design case in STR-001.

Case	Class S	Class M	Class L (PARAM-REF-MASS-OCC)	Class L (PARAM-REF-MASS-MAX)
Corner mass share	310–350 kg	450–550 kg	662.5 kg	787.5 kg
3 g operational per tower	8–11 kN	12–17 kN	20–23 kN	≈23 kN
20 g vertical per tower	60–70 kN	90–110 kN	≈130 kN	≈154.5 kN
×2 asymmetry per tower	120–140 kN	180–220 kN	≈260 kN	≈309 kN
Horizontal 1 g, total	12–14 kN	18–22 kN	≈26 kN	≈30.9 kN
Horizontal 20 g, total	250–280 kN	360–440 kN	≈520 kN	≈618 kN

All values STUDY. Risk R-08 compares crash loads with capture frame, stops and hardpoint capacity.

6.4 Stop-energy illustration. A placeholder of ≈15.6 kJ results from ½ m v² with m = PARAM-REF-MASS-OCC and v = 3.43 m/s (20 g over 30 mm free play). It is illustrative. STR-001 shall define the design energy event: initiating scenario, relative velocity, free-play geometry, corner versus whole-body share and mass case, and shall replace this value.

7. CONTROLLED REQUIREMENTS

7.1 Requirement table.

ID	Requirement	Owner	Gate	Status
REQ-M01	Upper body mechanically captured in all states; static capacity per class S/M/L at the §6.2 floors	Structures	G0/G2	BASELINE / TARGET
REQ-M02	Macro stroke PARAM-M02-MACRO-STROKE without clash	Mechanical	G1	TARGET
REQ-M03	Micro stroke PARAM-M03-MICRO-STROKE inside PARAM-LORENTZ-GAP envelope	Actuator	G1	TARGET
REQ-M04	Control bandwidth and carriage mode per PARAM-BW-CONTROL	Control	G2	TARGET
REQ-M05	Peak and RMS force per PARAM-FORCE-PEAK / PARAM-FORCE-RMS; Km measured, PARAM-ACT-KM-MIN stays ASSUMPTION until ACT-001	Actuator	G2	ASSUMPTION
REQ-M06	Core tower mass per PARAM-TOWER-MASS-LIMIT at PARAM-MASS-CONFIDENCE	Module	G2	TARGET
REQ-M07	Power-loss quantitative criteria (§7.2)	Safety	G2	BASELINE
REQ-M08	SIM-001 multi-band acceptance (§12.2)	Simulation	G1	TARGET
REQ-M09a	STR-001 per-class progressive-stop curve including §6.3 Class L cases	Structures	G3	TARGET
REQ-M09b	Stops absorb crash energy without the magnet loop; full curve (§7.6)	Structures	G3	TARGET
REQ-M10	Life per PARAM-LIFE	Reliability	G3	TARGET
REQ-M11	Operating temperature range frozen	Thermal	G2	OPEN
REQ-M12	EMI and cabin-field plan	EMC	G3	OPEN
REQ-M13	ICD-001 defines mechanical, electrical, communication, safety and lock interfaces	Systems	G1	TARGET
REQ-M14	No single failure causes loss of mechanical capture	Safety	G2	TARGET
REQ-M15	Two-step manual rescue under power and hydraulic loss (§7.3)	Safety	G3	TARGET
REQ-M16	If hold is fitted: rate-limited; not a primary crash path	Safety	G2	BASELINE (conditional)
REQ-M17	Required SPORT locks positively confirm engagement and release; time is an allocation inside PARAM-MODE-TIME (value TBD in ICD-001)	Lock	G2	TARGET
REQ-M18	COMFORT provides full macro envelope without lock interference	Mechanical	G2	TARGET
REQ-M19	Mode arbitration supports driver, radar/surface, GPS/map with driver priority	Controls	G3	STUDY
REQ-M20	No transition enters unconstrained free-float	Safety	G2	BASELINE
REQ-M21	SPORT↔COMFORT lateral-g policy (§7.4)	Dynamics	G2	TARGET
REQ-M22	Leveling preserves dynamic stroke and design attitude under payload change	Hydraulics	G2	TARGET
REQ-M23	Incomplete lock set inhibits SPORT declaration; lock state diagnosable and reported to ECU	Lock / Controls	G2	TARGET
REQ-M24	Fluid systems follow ISO 4413 safety path; interfaces in ICD-001	Hydraulics	G1	TARGET
REQ-M25	Horizontal load path defined and verified before integration-ready claim (§7.7)	Structures	G2	OPEN

PARAM-MODE-TIME is measured from accepted request to declared state (S2S or S2) and excludes REQ-M21 inhibit time.

7.2 REQ-M07 — Quantitative power loss. After loss of commanded active power the system shall: (1) detect it within a defined maximum detection time; (2) transition to S4; (3) close relevant valves within a defined maximum response time; (4) remain inside the mechanical capture envelope; (5) remain below defined maximum vertical displacement and velocity; (6) prevent uncontrolled free-float; (7) keep passive support at or above the applicable §6.2 floor. All limits are frozen in ICD-001 and verified at L2/L3. State definition alone is not acceptance.

7.3 REQ-M15 — Manual rescue. A mechanically available two-step rescue mechanism remains operable under the specified power-loss and hydraulic-failure conditions, does not depend on ECU aliveness, requires deliberate two-step activation and creates no unintended free-float. Interface, access location, force, sequence, indication, safe-state outcome and re-enable procedure are configuration-controlled in ICD-001 and verified by physical test.

7.4 REQ-M21 — Lateral-g policy (temporary until SIM-001, HIL and four-corner evidence).

Element	Rule
SPORT→COMFORT inhibit	Immediate when ay exceeds PARAM-M21-AY-HIGH, for automatic and manual requests; driver override cannot defeat it (a request may be queued only)
Re-enable	Only after ay stays below PARAM-M21-AY-LOW for PARAM-M21-HOLD
Filtering and latency	Defined and validated filter and latency budget, biased to inhibit; the former "no time filter" wording is SUPERSEDED
Sensor validity and faults	Validity criteria; defined safe arbitration on invalid, stuck or disagreeing sensors; no single lateral-acceleration sensor can create an unsafe transition
Transients	Noise handling; yaw and roll transients; braking and combined manoeuvres
COMFORT→SPORT	Also conditioned on lateral acceleration, yaw rate, roll rate and macro stroke position limits (parameters TBD under SIM-001/HIL) so seat-band drive and lock engagement under cornering load cannot cause unconstrained motion or lock damage
Roll gradient	PARAM-ROLL-GRADIENT-VL under SIM-001 / HIL
Manual rescue	Separate procedure (§7.3)

7.5 Sensor functional independence. Redundancy means true functional independence, not duplication. Evaluate common-cause failure through: common power supply, ECU, communication bus, software, mounting, environmental exposure, calibration error and vibration. A safety monitor independent of the primary control path shall be evaluated for critical mode arbitration and force-bound enforcement.

7.6 Progressive-stop curve. STR-001 defines a complete force-displacement and energy-absorption curve, not a single peak value: trigger displacement; initial engagement force; progressive force rise; peak force; absorbed energy; residual stroke; temperature sensitivity; manufacturing tolerance; repeatability; post-event inspection criteria. The stops absorb the design energy (§6.4) independently of Lorentz authority.

7.7 REQ-M25 — Horizontal load path. Closed before any integration-ready claim. Define and verify: longitudinal and lateral stiffness; yaw restraint; free play; load sharing between capture frame, links, stops and SPORT locks; braking, lateral, combined vertical/horizontal, crash and propulsion/brake torque-reaction loads; asymmetric four-corner loading. The Lorentz stage is never credited as primary crash or horizontal path.

8. STATE MACHINE

8.1 States.

State	Name	Definition
S0	Safe / Captured / Power-off	No powered release; capture remains
S1	Precharge / Self-test	Capture confirmed; 48 V path checked
S2	COMFORT isolation	Full envelope with mandatory capture
S2S	SPORT locked	All required locks confirmed; stages held

State	Name	Definition
S3	Degraded	Reduced authority; force bounded
S4	Power-loss capture	Passive support; valves fail-closed; quantitative envelope
S5	Impact preparation	Stops prepared; high-rate unlock inhibited
S6	Impact / arrest	Stops active; no magnet-energy dependence
S7	Post-event safe	No automatic re-enable
S8	Maintenance lockout	Physical lockout

8.2 Transitions.

From	To	Condition
S0	S1	Power-on self-test pass
S1	S0	Controlled shutdown
S1	S3	Self-test fail or critical fault
S1	S2	Driver or automatic COMFORT selection
S2	S2S	SPORT request; all required locks report engaged
S2S	S2	REQ-M21 permits; all locks report released
S2	S3	Critical fault, sensor invalidity, thermal derate, lock fault or accumulator pre-charge loss
S2S	S3	Same causes; locks are NOT released by this transition
S3	S2	Fault cleared per ICD-001 rules
S3	S1	Fault cleared and authorised reset
S2 / S2S	S5	Pre-crash sensor class and/or deceleration threshold (OPEN, ICD-001)
S5	S6	Measured impact or stop contact
S5	S2 / S2S	Impact-preparation abort
S6	S7	Event handling complete
S7	S1	Manual clear by authorised personnel only
S1 / S2 / S2S	S8	Physical lockout, only while stationary
S8	S1	Lockout removed by physical key / procedure
Any powered state	S4	Power loss; valves fail-closed; engaged locks spring-applied fail-locked
S4	S1	Power restored and self-test pass (only exit from S4)

S0 is excluded from "Any powered state → S4".

9. SAFETY ANALYSIS

9.1 Screening FMEA. S, O, D use a 1–10 screening scale (inherited, not evidence-based); RPN = S×O×D. SAF-001 replaces these with formal analysis.

Function / failure	Effect	Mitigation	S	O	D	RPN
Carriage jam	Loss of micro authority	Sensors, stops, passive support	9	3	4	108
Link fracture	Local free-motion risk	Capture frame, stops	9	2	5	90
Leveling stuck	Stroke consumed	Level sensor, S3	8	3	3	72
Hose burst	Loss of support/leveling	Pressure monitoring, stops	9	3	3	81
Coil overheat	Derate / shutdown	Temperature model	7	3	3	63
Magnet force-constant loss	Force drop	Temperature limits, derate	7	2	4	56
SPORT lock fails to engage	Remains COMFORT	Status sensors, alert	6	3	2	36
Partial lock (1–2 of 4)	Asymmetric stiffness	Inhibit S2S	8	2	2	32
False COMFORT at high lateral g	Excess roll	Lateral-g / yaw inhibit	7	2	3	42
Horizontal path overload	Capture shear exceeded	REQ-M25, stops	9	2	4	72
Power loss	Active force lost	S4 with §7.2 limits	9	SAF-001	SAF-001	—
Software / OTA fault	Bad command	Bounds, S3/S4	8	3	4	96
Unintended SPORT lock at speed	Handling/comfort conflict	Arbitration inhibit, alert, R-17	8	3	3	72
Accumulator pre-charge loss	Macro response degraded	Pressure sensing, S3, R-18	7	3	3	63
Hydraulic leak / fire	Loss of support, fire risk	Fluid selection, containment, R-19	9	2	4	72
Common-cause sensor fault	False mode or force	Functional independence and independent safety monitor (§7.5)	8	3	3	72
Magnet demagnetisation	Force-constant drop	Temperature limits, derate, R-21	7	2	4	56

9.2 Preliminary HARA (screening; severity class is written Sev-class to avoid confusion with FMEA S).

Hazard	Scenario	Sev-class / E / C	Draft safety goal
Loss of capture	Carriage or link failure	S3 / E3 / C3	Maintain capture under single failure
Uncontrolled vertical motion	Force runaway	S2 / E3 / C2	Bound force; S3 or S4
Asymmetric SPORT lock	Partial lock plus lateral load	S2 / E3 / C2	Inhibit S2S unless complete
Horizontal overload	Brake or crash without defined path	S3 / E2 / C3	Size paths under REQ-M25
Unexpected height drop	Valve open at speed	S2 / E3 / C2	Fail-closed valves
Post-crash automatic re-enable	Uncommanded motion	S2 / E2 / C2	S7, no automatic re-enable
Unintended SPORT lock	False engage on rough road	S2 / E3 / C2	Inhibit, alert, track under R-17
Unlock under high lateral g	Release request in a turn	S2 / E3 / C2	REQ-M21

9.3 Safety process. No ASIL is assigned and no safety case is claimed. SAF-001 evolves into a configuration-controlled safety case: HARA, FMEA, FTA, common-cause and dependent-failure analysis, fault injection, safe-state definition, diagnostic coverage, single-point and latent-fault analysis, verification traceability. Mechanical capture, passive support and progressive stops are independent of E/E and form the primary safety backbone. Cyber and OTA paths able to change force bounds or mode policy require authenticity and integrity controls. Candidate references are in Annex B; no compliance is claimed.

10. ACTUATOR MAP, GAP ENVELOPE AND THERMAL

10.1 Km evidence chain. PARAM-ACT-KM-MIN stays an assumption until validated by measured force-current-power-gap data. The verified ACT-001 map, not nominal Km, is the design input for control and thermal sizing.

Axis	Sweep / points	Record
Air gap	Across the full tolerance envelope, min to max	Force, current, power
Temperature	Cold, nominal, hot	Force, resistance, magnet temperature
Current	RMS region to peak	Force, power, thermal rise
Duty cycle	Continuous and pulsed	Thermal equilibrium
Manufacturing sample	Minimum agreed number of samples	Spread and capability
End-of-life surrogate	If available	Derating note

10.2 Gap tolerance envelope. PARAM-LORENTZ-GAP is a complete tolerance-controlled envelope, not a nominal range. ACT-001 demonstrates force capability and safe control authority across the validated envelope.

Worksheet field	Purpose
Nominal / minimum / maximum gap	Design centre; contact-risk bound; authority bound
Assembly tolerance	Build variation
Structural deflection under load	Operating shift
Thermal growth	Hot/cold shift
Vibration-induced motion	Dynamic envelope
Manufacturing tolerance	Part capability
Sensor error	Control uncertainty
Stop-to-gap relationship	End-of-travel coordination

10.3 Thermal. Continuous thermal design is dominated by copper loss at the RMS operating point; continuous and transient limits are separated. Continuous RMS capability and REQ-M11 stay OPEN until bench evidence closes the model. Demagnetisation (R-21) is evaluated across temperature and duty cycle. SPORT copper demand is near zero (stages held; evidence OPEN).

11. MODE SEQUENCES AND LOCK ARCHITECTURE

11.1 COMFORT→SPORT.

Step	Action	Interlock
1	Arbitration accepts request	No active lateral-g or fault inhibit; §7.4 COMFORT→SPORT conditions met
2	Macro and leveling drive body to seat band	Stroke sensors in range
3	Seat contact detected at required corners	Contact flags true
4	Positive locks command engage	Actuator enable healthy
5	All required locks report engaged	Incomplete set aborts to COMFORT
6	Stages held; state S2S	Driver indication ON

SPORT is never declared while a required lock is disengaged.

11.2 SPORT→COMFORT. Permitted only per §7.4. Locks command release; all report released; macro envelope re-enabled; micro control re-enabled; state S2. COMFORT is never declared while a required lock is engaged.

11.3 Lock architecture (LCK-001).

Element	Function	Status
Seat faces	Upper hardpoint to lower receiver at seat band	Geometry OPEN
Positive lock	Pin, claw or hydraulic clamp preventing separation	Per-corner status required
Actuation	Hydraulic or electromechanical	Duty cycle STUDY
Fail behaviour	Known state with driver alert; never free-float	BASELINE
Crash role	May share load; progressive stops remain primary energy path	BASELINE

12. VERIFICATION FRAMEWORK

12.1 Evidence ladder.

Level	Scope	Status
L0	Architecture and requirements (this document)	BASELINE
L1	SIM-001 numerical run with sensitivity and combined manoeuvres	OPEN
L2	Single-tower bench: force-gap-current, thermal RMS, power loss, lock cycle, stop contact	OPEN
L3	Four-corner / HIL: asymmetry, mode transition, stops, fault injection	OPEN
L4	Optional mule: ISO 2631-1 and handling on donor chassis	OPTION

12.2 SIM-001 specification.

Item	Definition
Model class	Quarter-car / half-car: upper mass, macro k and c, optional micro force; SPORT as high-k constraint
Reference passive	Frozen passive hydro-pneumatic k and c stated on every plot
Roads	ISO 8608 segments, at least one rough and one smooth
Mass states	Empty, occupied, maximum (Annex X mass basis = §6.2 / §6.3 references)
Metrics	aw (0.5–80 Hz), VDV, MSDV (0.1–0.5 Hz); axes, weightings (Wk, Wd; Wf for MSDV) and measurement point (seat surface or floor) stated per CMFSB-TP-001
Outputs	Transmissibility, band RMS, stroke histogram, roll proxy SPORT vs COMFORT, transition time history
SIM-C1	COMFORT lower RMS than frozen passive in at least one agreed band on a rough profile
SIM-C2	SPORT higher roll stiffness (lower roll) than COMFORT on the same lateral step
SIM-C3	Macro within stroke without continuous stop hammering
SIM-C4	No unconstrained free-float interval in transition cases

Results are PRELIMINARY until independent review.

12.3 Power-loss test (REQ-M07). Numeric limits are those frozen in ICD-001; until then results are developmental only.

Step	Action	Record
1	Establish COMFORT at defined payload and temperature	State, payload, temperature
2	Remove commanded active power (controlled cut)	Method, timestamp
3	Measure detection time	ms
4	Measure valve closure time	ms

Step	Action	Record
5	Measure peak vertical displacement and velocity	mm, mm/s
6	Confirm final settle inside capture envelope	Pass/fail vs ICD
7	Confirm no uncontrolled free-float	Pass/fail
8	Confirm passive support adequacy for load class	Pass/fail
9	Repeat from SPORT (S2S)	Full record set
10	Brown-out cases as agreed	Developmental

12.4 Lateral-g interlock test (REQ-M21).

Case	Stimulus	Expected
Threshold inhibit	ay steps above PARAM-M21-AY-HIGH	SPORT→COMFORT inhibited
Hysteresis re-enable	ay below PARAM-M21-AY-LOW for PARAM-M21-HOLD	Release permitted if other conditions OK
Single sensor fault	One path invalid or stuck	No unsafe release
Redundant disagreement	Paths disagree beyond limit	Safe inhibit or degraded rule
Yaw/roll transient	Combined dynamics profile	No nuisance or unsafe transition
Braking combined	Longitudinal + lateral	Policy-compliant behaviour
Driver override attempt	Override during inhibit	Cannot defeat interlock
COMFORT→SPORT in a turn	Request above the §7.4 conditions	Request held or refused; no free motion

12.5 Progressive-stop inspection (post-event, STR-001).

Check	Criterion	Disposition
Visible deformation	Within STR-001 allowables	Go / no-go for reuse
Residual stroke	Minimum residual retained	Go / no-go
Attachment integrity	No critical fastener or parent failure	Repair / replace
Force-displacement sample	Within curve tolerance if sampled	Accept / investigate
Temperature exposure	Record if extreme	Engineering review
Repeat-event policy	Per STR-001	Replace if required

12.6 Evidence record and reference protocol. Every record retains software version, parameter set, configuration hash, hardware configuration, road/profile ID, payload, mode, passive k and c, time histories, sensor calibration, uncertainty and data-quality flags, and review status. CMFSB-TP-001 defines: vehicle mass; payload; road profile; speed; tyre pressure; passive baseline; COMFORT; SPORT; sensor locations; sampling rate; filtering; metrics; acceptance criteria; uncertainty; configuration hash. Recommended demonstrator: one tower rig showing COMFORT, SPORT lock, power cut to S4, and sensor/actuator fault injection to a safe degraded state.

13. GATES, PHASES, RISKS, WORK PACKAGES AND OPEN ITEMS

13.1 Gates.

Gate	Intent	Entry evidence
G0	Concept controlled	Document issued; legal owner recorded
G1	Interface and simulation ready	ICD-001 draft; SIM assumptions frozen; clash check; Packaging Gate status recorded; REQ-M07 limit placeholders; reviewer appointed
G2	Single-tower evidence	Force, thermal, power-loss, lock bench; ACT-001 map start; REQ-M07 limits frozen
G3	Four-corner / HIL	Mode transitions; stop curve; fault injection; REQ-M25 path
G4	Optional mule	ISO 2631-1 and handling on donor
G5	Licence / transaction readiness	Buyer-accepted package; commercial term sheet; IP filing status; ownership chain

13.2 Phases (planning; no commitment without funding).

Phase	Month	Content
Ph0	0–2	CAD envelope, lock faces, ICD-001 draft; procurement package after ICD baseline
Ph0b	2–6	Supplier lead time (4–8 months planned), COTS surrogate bench, SIM-001 run; 90-day Option deliverables limited to SIM-001 and ICD-001 draft
Ph1	6–10	Single-tower force, thermal, power loss, lock cycle; ACT-001 map start
Ph2	10–15	Four-corner 6-DOF; mode transitions; stop curve; power loss from each mode; REQ-M25 path
Ph3	15–21 (optional)	Donor-chassis ISO 2631-1 and handling

13.3 Risk register (L and I on a 1–5 scale).

ID	Risk	L	I	L×I	Owner
R-01	Core tower mass exceeds limit at PARAM-MASS-CONFIDENCE	4	4	16	Module
R-02	Km not achieved / not demonstrated by measured map	3	5	15	Actuator
R-03	Lorentz thermal or force-gap limits	3	5	15	Actuator
R-04	Macro–micro clash or carriage failure	3	4	12	Mechanical
R-05	Stroke consumed without leveling	3	4	12	Hydraulics
R-06	Regulatory path unclear	3	4	12	Homologation
R-07	Vehicle P90 exceeds host mass ceiling	4	4	16	Mass
R-08	Per-tower crash load versus capture frame, stops and hardpoints (§6.3)	4	5	20	Structures
R-09	Comfort unproven until L1/L2	4	4	16	Simulation / PoC
R-10	EMI or field-exposure cost	2	4	8	EMC
R-11	Hydraulic mass above STUDY value	3	3	9	Hydraulics
R-12	Liability if safety case or claims exceed evidence	2	5	10	Legal / Safety
R-13	SPORT lock reliability or partial engagement	3	4	12	Lock
R-14	Mode-arbitration false positive or negative	3	3	9	Controls
R-15	Horizontal restraint undersized (REQ-M25)	3	5	15	Structures
R-16	Clearance-stack energy underestimated	3	4	12	Structures
R-17	Unintended SPORT lock at speed on rough road	3	4	12	Systems

ID	Risk	L	I	LxI	Owner
R-18	Accumulator pre-charge loss	3	3	9	Hydraulics
R-19	Hydraulic leak or fire	2	5	10	Hydraulics
R-20	Common-cause sensor fault	3	4	12	Systems
R-21	Magnet demagnetisation	2	4	8	ACT-001
R-22	Packaging residual remains negative at engagement	4	4	16	Packaging Gate
R-23	IP filing delayed relative to external disclosure	3	5	15	IP-001

R-08 is the highest screening score.

13.4 Work packages.

Priority	Work package	Deliverable
G1	Freeze CAD envelope, lock faces, leveling, fail-closed valves; Packaging Gate status	3-D package, clash report, residual status
G1	ICD-001 draft (mechanical, electrical, communication, safety, lock) with REQ-M07 placeholders	Buyer-readable ICD set
G1	SIM-001 assumptions and first COMFORT-vs-passive run	Plot pack v0.1 and configuration hash
G2	Single-tower force, thermal, power loss, lock cycle	Bench data package
G2	ACT-001 map start; STR-001 curve; REQ-M25 definition	Maps and structures memo
G2	Formal HARA, FTA, common-cause analysis	SAF-001 work products
G3	Four-corner mode transitions and fault injection	6-DOF records and transition-time records
G5	Licence / IP-transaction package; IP filing status	Commercial pack under counsel (COM-001)

13.5 Document register.

ID	Scope	Status	Gate
DOC-00	This baseline (R5.16)	Issued	G0
MOD-001	Module datasheet (1–3 page buyer summary)	To issue	G0
ICD-001	Interface control	OPEN	G1
SIM-001	Multi-band simulation	OPEN	G1
CFG-001	Configuration management and review disposition	OPEN	G1
HYD-001	Hydraulic macro stage, valves, leveling	OPEN	G1/G2
STR-001	Structure, progressive stops, crash path, REQ-M25	OPEN	G2
LCK-001	Lock system	OPEN	G2
ACT-001	Actuator map, thermal, efficiency	OPEN	G2
ENE-001	Power and current envelope	OPEN	G2
SAF-001	HARA, FMEA, FTA, common-cause, safety case	OPEN	G2
TST-001/002	L2 / L3 test plans	OPEN	G2/G3
BRK-001	Brake and horizontal-load interface	OPEN	G3
REG-001	Regulatory pathway	OPEN	G3
IP-001	IP family, prior art, claim chart, ownership, FTO	OPEN	Counsel
COM-001	Commercial terms (T3)	Counsel-controlled	Option

13.6 Open-item tracker.

ID	Item	Owner class	Gate
O-01	Packaging residual V-L non-negative with all §4.2 lines	Packaging	G1
O-02	REQ-M07 numeric limits in ICD-001	Safety / Systems	G2
O-03	REQ-M15 physical trial under power and hydraulic loss	Safety	G3
O-04	REQ-M21 full policy evidence and PARAM-M21-HOLD	Dynamics	G2
O-05	REQ-M25 horizontal path closure	Structures	G2
O-06	STR-001 curve and design energy event	Structures	G3
O-07	ACT-001 actuator map	Actuator	G2
O-08	ENE-001 total power and current	Electrical	G2
O-09	Formal HARA / FTA	Safety	G2
O-10	Patent filing number and date	IP / Counsel	G5
O-11	Ownership chain and entity assignment records	Legal	G5
O-12	Independent reviewer appointment (PARAM-REVIEWER-DEADLINE)	Programme	G1
O-13	REQ-M11 temperature range	Thermal	G2
O-14	REQ-M12 EMC / cabin field	EMC	G3
O-15	PARAM-MASS-CONFIDENCE and aggregate P90 method	Mass	G1
O-16	Mode-time allocation (lock time, inhibit exclusion)	Controls / Lock	G2

14. HOST VARIANTS (PATH B)

Host variants are application-layer references. Core module IP does not depend on V8, V12, 4×4, 6×6 or CFRP, and Path A does not require any host figure. LOCKED means engagement decision, not verification, not packaging closure; the Packaging Gate (§4) applies. Engine entries are class descriptors, not brands.

14.1 Geometry and powertrain.

Parameter	V-L Urban Light	V-H Military / Off-road	Status
Length	5,000 mm	5,500 mm	LOCKED
Width (mirrors excluded)	2,000 mm	2,200 mm	LOCKED
Height (SPORT / seated)	1,700 mm	2,000 mm	LOCKED
Wheelbase	3,000 mm	3,300 mm (front axle to rear-tandem midpoint)	LOCKED
Ground clearance (SPORT)	200 mm	300 mm	LOCKED
Drive	4×4	6×6 (front axle + dual rear tandem)	LOCKED
Design occupants	4	6	LOCKED
Passenger cell	Full CFRP structural cell	Full CFRP structural cell	BASELINE

Parameter	V-L Urban Light	V-H Military / Off-road	Status
Roll structure	FIA-referenced CFRP/hybrid (no homologation)	Same	Reference only
GVW ceiling	PARAM-GVW-VL	PARAM-GVW-VH	LOCKED
Powertrain	Hybrid ICE + e-drive	Hybrid ICE + e-drive	LOCKED
ICE class	V8, 4.0 L twin-turbo class	V12, 6.0 L twin-turbo class	LOCKED class
ICE power	≈430–450 kW class	≈460–500 kW class	TARGET
Electric traction	≈150–180 kW	≈200–250 kW	TARGET
Battery usable / fuel tank	20 kWh / 80 L	40 kWh / 150 L	LOCKED
Range	1,000 km	700–800 km	STUDY (recompute road load at 90/120 km/h in ENE-001; V-L ≈8 L/100 km is optimistic)

14.2 Carbon cell and roll structure. Conventional CFRP layup and bonded construction is baseline until continuous-fibre printing is evidenced. Cell mass budget ≤280 kg (V-L) and ≤420 kg (V-H), TARGET. Roll-structure mass budget ≤45 kg and ≤70 kg. Roll-structure philosophy may reference FIA Appendix J concepts subject to verification of article numbers and edition; load cases are OPEN until edition freeze.

14.3 Class mapping and integration. V-L maps to Class S; V-H to Class M/L. A 4×4 pilot is generally the shorter first-contract path; 6×6 is a funded second SKU with higher axle, steering and packaging risk, optional on V-H only if axle mass is traded from reserve or protection. Hardpoints match host chassis stiffness; brake and propulsion torque reactions shall not overload horizontal links.

14.4 Closed mass budgets (sole valid host budget; STUDY).

WBS line	V-L kg	V-H kg
CFRP passenger cell	280	420
Roll members	45	70
Closures, glass, panels / reinforced exterior	230	400
Interior, HVAC, cabin LV	200	330
Lower chassis	340	720
ICE and ancillaries	275	390
Hybrid transmission / coupling	160	190
E-motors and inverters	130	180
Battery (usable energy per §14.1)	145	260
HV/LV and cooling	95	160
Axles, steering / three-axle driveline	260	580
Wheels, tyres, brakes	210	480
Isolation, four corners (Class S / M-L)	280	380
Optional light protection package	—	200
Fuel tank and reference fuel	75	140
Exhaust and fluids	70	100
Programme reserve	70	150
Curb mass, no occupants	≈2,865	≈5,150
Occupants (PARAM-OCC-MASS × 4 / × 6)	400	600
Cargo / mission payload	100	400
GVW design case	≈3,365	≈6,150
Margin to ceiling	≈635	≈850

V-H includes the optional 200 kg protection package; if omitted it moves to reserve. REQ-V01/V02: GVW claims stay within the ceilings including the occupant policy. REQ-V03: both variants use a CFRP cell within these budgets. REQ-V04: FIA-referenced roll structure without homologation claim. REQ-V05: external materials name V-L and/or V-H whenever mass is quoted.

15. SUPPLIER AND ICD OUTLINE

15.1 Supplier data requests (no supplier is committed by this document; catalogue values are not programme evidence).

#	Domain	Request
1	Lorentz	Force-current-gap-temperature maps, or commitment to support ACT-001; continuous and peak force; gap tolerance; thermal derating curves
2	All	Mass P50/P90 with process-capability statement
3	Hydraulics	Cylinder force at pressure; seal life; fail-closed valve response and leakage class; duty, contamination and service intervals; hold-option capability
4	Locks	Engage/release time under load; status sensing; shear capacity; contamination tolerance
5	Stops	Force-displacement curve; energy rating; temperature sensitivity; inspection interval
6	Sensors	Stroke, force, pressure, temperature accuracy and fault behaviour; independence architecture
7	Electronics	48 V drive capability; current limit; diagnostic coverage; EMC data or test plan
8	Commercial	Lead time; change control; IP-clause willingness; quality system and traceability level

15.2 ICD-001 minimum chapters. Hardpoint geometry and loads; hydraulic ports and fluid; electrical power and signals (48 V class with full current envelope); mode and interlock logic; progressive-stop curve reference; REQ-M07 limits; thermal limits; installation torque and sequence; maintenance and rescue release; service access; EMC and grounding; configuration identity.

16. TRANSACTION READINESS (PATHS A, B, C)

16.1 Transaction paths.

Path	Object	Typical inclusions	Not included unless separately agreed
A — Licence / supply	Right to use the CMFSB module architecture in a named field	DOC-00 and agreed controlled subset; ICD, datasheet, safety behaviour, dual-mode rules; agreed engineering support hours; updates within a defined revision window; Option milestones	Assignment of all IP; certified vehicle or type approval; unlimited R&D; production tooling; guaranteed comfort percentage; guaranteed regulatory approval
B — Vehicle programme	Module plus host variant V-L and/or V-H	Host packs (§14)	Vehicle certification packs, series production

Path	Object	Typical inclusions	Not included unless separately agreed
C — IP assignment	Assignment of the module IP family	Bilateral terms subject to counsel, filing status and evidence gates	Anything not in the assignment schedule

Default Path A Option: non-exclusive evaluation in a named field; limited exclusivity as a paid add-on. Contracting party is Amirbahador Atai (individual) until formal assignment to a legal entity; the entity path (owner, target date) is recorded under IP-001 and COM-001 before any Option is signed, and liability allocation is defined in COM-001.

16.2 Readiness boundary.

Use case	Suitable on this document	Condition
NDA and T1 diligence	Yes	Under NDA
Paid Option negotiation	Yes	Bilateral term sheet in COM-001
Module licence term sheet	Conditional	ICD-001 draft plus SIM-001 or a committed bench plan
Large upfront assignment or exclusive sale	Not on this document alone	IP filing status, ownership chain, counsel FTO review, evidence gates
Production or type-approval claim	No	Non-claims §1.2
Advertising measured ride comfort	No	Until L1/L2 configuration-managed results

16.3 Buyer diligence matrix.

Buyer question	Current answer	Evidence / gate
Is the architecture defined and safe by construction?	Mechanical capture, passive support and independent stops are architectural invariants (§2.3)	SAF-001, G2/G3
Does it work?	Not demonstrated; TRL 2–3	SIM-001 (L1), bench (L2)
Is the actuator real?	Km is an assumption	ACT-001 map
Does it fit?	V-L stack negative; V-H +10 mm before missing lines	Packaging Gate, O-01
What does it weigh?	Core tower 48 kg P50; 67 kg line-sum P90 vs limit	O-15, supplier data
Who owns it?	Amirbahador Atai (individual)	IP-001 ownership chain, O-11
Is it protected?	Filing status per Annex T3-A.4	O-10
What is the safety status?	Screening only; no ASIL	SAF-001

16.4 Buyer information set (minimum for a serious discussion under NDA).

Element	Source	Tier
This baseline	DOC-00 R5.16	T1
Module datasheet extract	MOD-001	T1
Safety behaviour description (S4, locks, stops)	§2, §7, §8	T1
ICD draft or outline	ICD-001	T1/T2
SIM-001 assumptions list	SIM-001	T1/T2
Open-item list	§13.6	T1
IP overview without claim charts	§16.5	T1
IP schedule, claim charts, prior-art mapping	Annex T3-A	T3
Commercial term sheet	COM-001	T3

16.5 IP statement for T1 distribution. Multi-family IP protection is in preparation under counsel. No statement of patent novelty, enforceability or freedom-to-operate is made. IP protection proceeds in parallel with engineering and does not wait for prototype completion. No T1 document is released to a third party before a provisional filing or written counsel clearance.

16.6 Conditions precedent to any assignment or exclusive licence (checklist).

#	Condition	Evidence
1	Provisional (or later) filing(s) made and recorded	Application number, office, date
2	Inventor declarations and chronology complete	IP-001
3	Assignments and contractor / supplier IP clauses documented	Ownership chain file
4	Confidentiality and disclosure history documented	NDA register
5	Prior-art search and claim chart reviewed by counsel	Annex T3-A
6	FTO review by qualified counsel	Counsel opinion
7	Independent technical reviewer appointed and findings dispositioned	CFG-001
8	Entity assignment path defined	IP-001 / COM-001

17. GOVERNANCE AND SIGN-OFF

17.1 Configuration control. One controlled source per numerical parameter (Section 3); every critical parameter has a unique ID; historical values are retained only as SUPERSEDED. CFG-001 minimum content for each controlled change:

Item	Required content
Document revision	ID, date, supersedes
Parameter change	Old value, new value, reason
SUPERSEDED mark	Prior value retained for traceability
OPEN-item closure	Evidence reference and gate
Review finding	Disposition and owner
Evidence configuration hash	Software, parameters, hardware serial class
Disclosure tier	T0–T3

DOC-00 is the narrative baseline; design freeze occurs in ICD-001 and part drawings, not in DOC-00 alone. All TARGET and STUDY values that become design inputs are configuration-managed.

17.2 Independent reviewer challenge list (before G1 credit).

#	Challenge topic
1	Capture claims versus free-levitation misunderstanding
2	Mass P50 versus confidence-level consistency
3	Km assumption exposure in external materials
4	Static floor rule and mass-basis consistency

#	Challenge topic
5	Packaging residual versus LOCKED language
6	REQ-M07 measurability (limits frozen?)
7	REQ-M15 operability under dual failure
8	REQ-M21 single-sensor and disagreement behaviour
9	REQ-M25 openness versus any integration-ready language
10	Progressive-stop curve completeness and design energy event
11	IP ownership chain completeness
12	Configuration discipline and one-value-one-place
13	State-machine completeness (S3 entry, S4 exit)

17.3 Release and sign-off.

Role	Name / status	Date / target
Legal Owner / Author	Amirbahador Atai	2026-10-06
Independent Technical Reviewer	TBD	PARAM-REVIEWER-DEADLINE
Approval Authority	TBD after independent review	After G1 evidence
Documentation authority	ATFSV Project Engineering Documentation Authority	2026-10-06

Status: CONTROLLED / Contract Diligence Baseline. Not approved for road use, production or certification credit. All TARGET, STUDY, ASSUMPTION, DERIVED, LOCKED and OPTION values require formal verification before design release. Priority before serious OEM or investor engagement: (1) ACT-001, STR-001 and SIM-001 evidence; (2) SAF-001 including quantitative power-loss limits, sensor independence and horizontal load path; (3) IP-001 filing, prior art, claim chart and ownership chain.

T3 — COUNSEL CONTROLLED

ANNEX T3-A — IP SCHEDULE (T3 — COUNSEL CONTROLLED — REMOVE FROM ALL T1/T2 DISTRIBUTION)

T3-A.1 Proposed patent families (for counsel review).

Family	Scope	Treatment
A	Core mechanically captured macro–micro isolation architecture	Patent candidate
B	Hybrid hydraulic macro + high-bandwidth micro actuator architecture	Patent candidate
C	Positive SPORT lock and mechanical capture geometry	Patent candidate
D	Safety state machine; S4 passive-support with quantitative envelope; transition constraints	Patent / trade-secret review
E	Dynamic mode arbitration and control; REQ-M21 interlock logic	Patent / software-protection review
F	Progressive-stop and independent crash path with full force-displacement absorption	Patent candidate
G	Modular host-class integration interface (four-tower ICD, load sharing)	Patent candidate
Trade secrets	Actuator tuning, calibration maps, control parameters, manufacturing processes, supplier-specific implementation	Trade-secret candidate

T3-A.2 Invention-disclosure candidates (subject to counsel): (1) mechanically captured macro–micro isolation architecture; (2) passive mechanical support combined with high-bandwidth Lorentz micro-actuation; (3) dual COMFORT/SPORT operating architecture; (4) positive mechanical SPORT lock architecture; (5) safety state machine and transition constraints; (6) S4 passive-support architecture with quantitative envelope; (7) progressive-stop and independent crash-load path with full force-displacement curve; (8) horizontal-load restraint architecture; (9) integration interface across multiple host classes; (10) coordinated control method combining vehicle dynamics, isolation state and mechanical locking.

T3-A.3 Four-layer IP structure and differentiation matrix.

Layer	Content	Purpose
1 Core architecture	Capture + macro isolation + micro actuation + stops	Survives change of actuator technology
2 Dynamic behaviour	COMFORT/SPORT, transition logic, lateral-g policy	Product differentiator
3 Safety architecture	S4, independent crash path, fail-closed, positive lock	OEM safety-case relevance
4 Integration platform	Four-tower interface, loads, modularity	Licensing across host classes

Feature	Prior art	What is different	Patent candidate	Trade secret
Mechanical capture	TBD by counsel	Always kinematically constrained; no free levitation	Yes	No
Macro–micro isolation	TBD by counsel	Hybrid hydraulic + high-bandwidth micro stage inside mechanical envelope	Yes	Tuning, maps
SPORT lock	TBD by counsel	Positive mechanical lock for handling stiffness	Yes	No
S4 power loss	TBD by counsel	Fail-closed + passive support + quantitative envelope	Yes	No
REQ-M21 logic	TBD by counsel	Lateral-g interlock with full sensor, fault and transient policy	Yes	Parameters
Progressive stop	TBD by counsel	Independent of magnetic closed loop; full curve	Yes	No
Module interface	TBD by counsel	Host-class adaptable four-tower ICD	Yes	No

T3-A.4 Filing and ownership record (to be completed and dated by counsel).

Item	Record
Provisional application(s)	None filed as of 2026-10-05 per project record. Update: No. [], Office [], Date []
Inventor(s)	Amirbahador Atai [confirm complete list and chronology]
Current owner	Amirbahador Atai (individual)
Assignment to entity	[Entity, date, instrument — TBD]
Contractor / supplier IP clauses	[TBD]
Confidentiality and disclosure history	[NDA register — TBD]
Jurisdiction strategy	[TBD by counsel]
FTO opinion	[TBD by counsel]

IP-001 content: invention disclosure, inventor chronology, prior-art search, patent landscape, claim chart, potentially conflicting families, jurisdiction strategy, ownership chain, assignment records, confidentiality history, FTO review.

T3-A.5 Term-sheet topic schedule (topics only; values are bilateral and live in COM-001).

Field of use · territory · exclusivity and its fee · Option period and exercise mechanics · milestone and royalty structure · improvements and grant-back · ownership and title warranties · IP representations and indemnity · data-room and escrow arrangements · supplier flow-down · audit rights · export-control compliance · confidentiality and publication · termination and reversion · governing law and dispute resolution.

ANNEX B — REFERENCES AND GLOSSARY

Reference	Use	Status
ISO 26262	Functional-safety process	Candidate; compliance not claimed
ISO 21448	Safety of the intended functionality	Candidate
ISO 2631-1	Whole-body vibration evaluation	Candidate
ISO 8608	Road-surface profiles for simulation	Candidate
ISO 4413	Hydraulic fluid power safety (REQ-M24)	Reference
ISO 13849-1	Lock and valve safety functions if used	Candidate
UN R100 / ISO 6469-3	HV safety if applicable	Candidate
UN R94 / R95 / R14 / R16	Crash and restraint matters if applicable	Candidate
EN 1789 / EN 1865	Ambulance planning references (Application A only; no clinical claim)	Candidate
FIA Appendix J	Roll-structure philosophy only	Reference

Glossary: CMFSB — Captured Mechanical Floating Safety Body. S4 — power-loss passive-support state. COMFORT / SPORT — dual running-gear modes (S2 / S2S). Application A — ambulance and clinical-transport planning context. Path A / B / C — licence, vehicle programme, IP assignment. VDV — vibration dose value. MSDV — motion sickness dose value. P50 / P90 — median and 90th-percentile planning values.

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